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Approaching Sustainable Last-Mile Delivery: What Institutional Pressures Shape Adoption by E-Commerce Companies in Urban Contexts

A qualitative study through the lens of Institutional Theory

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Abstract

Background: The rapid growth of e-commerce has intensified last-mile delivery operations in urban areas, making them a significant source of carbon emissions. While sustainable delivery solutions such as electric vehicles, cargo bikes and relay-point networks are increasingly available, the existing literature remains predominantly quantitative, focused on optimisation models, and often overlooks the organisational perspective of e-commerce companies navigating this transition.

Purpose: The purpose of this study is to explore what institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies in urban contexts.

Method: This study follows a qualitative research design using an abductive approach. Eleven semi-structured interviews were conducted with professionals from e-commerce companies of varying sizes operating in urban contexts.

Conclusion: The findings reveal that normative pressures are the dominant institutional force shaping sustainable last-mile adoption across the sample. Coercive pressures, such as government regulation, operate indirectly through anticipatory compliance, carrier-level delegation and urban infrastructure constraints rather than through direct regulatory enforcement. Mimetic pressures are the weakest mechanism, partly due to the fragmented and niche nature of the industries studied. These findings suggest that, in the contexts studied, enabling urban infrastructure and normative signalling may complement regulatory mandates as levers for accelerating sustainable delivery adoption.

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1. Introduction

This section provides an overview of the increasing environmental challenges regarding transportation in logistics and thus the importance of implementing sustainable solutions. It also outlines the research problem, purpose, perspective and delimitations.

1.1 Background

Transportation plays a central role in logistics management. It enables the movement of goods between suppliers, manufacturers, distribution centres and final consumers. Over the past decades, the expansion of global trade and the increasing complexity of logistics systems have significantly intensified cargo transport activities. Nowadays, goods circulate across continents through large logistics networks relying on several modes of transport. These modes mostly include road, rail, maritime shipping and air freight (Ritchie & Roser, 2021). Each fulfils a specific role: road transport dominates short and medium distance deliveries due to its flexibility and door-to-door capabilities. Rail and maritime transport are typically used for transporting large volumes over longer distances. Air cargo, although more expensive, is generally reserved for high-value or time-sensitive shipments.

Alongside the globalisation of trade, the rapid expansion of e-commerce has deeply transformed logistics operations. Over the past two decades, online retail has experienced significant growth worldwide. Companies such as Amazon, Alibaba and many other digital retailers have reshaped consumer expectations and purchasing behaviour. Same-day or next-day delivery has become increasingly common because consumers demand fast, reliable, flexible delivery services. As a result, e-commerce companies have had to redesign their logistics networks, develop new distribution infrastructures and increase transportation capacity to meet these expectations.

This transformation has significantly intensified logistics activities, particularly in urban areas. The rise of last-mile delivery, the final stage of the delivery process from distribution centres to end consumers, has emerged as one of the most operationally complex and carbon-intensive segments of modern logistics (Savelsbergh & Van Woensel, 2016). Urban environments introduce structural constraints, including traffic congestion, limited access windows and stringent local emission regulations. Those barriers fundamentally modify how delivery

operations are planned and evaluated. E-commerce companies rely a lot on road transportation to ensure quick deliveries, which has led to an increase in urban freight demand. While these developments have improved service efficiency and customer satisfaction, they have also contributed to growing environmental concerns.

Indeed, the continuous growth of cargo transportation has made the transport sector one of the largest contributors to greenhouse gas emissions worldwide. In 2019, transport activities generated approximately 8.7 gigatons of CO_2 equivalent emissions, representing around 23% of global energy-related CO_2 emissions (IPCC, 2022). Road transportation alone accounts for nearly 70% of these emissions. Logistics transport and warehousing activities, together, represent approximately 7% of global greenhouse gas emissions (McKinsey & Company, 2024).

These environmental challenges have increased pressure on both companies and governments to reduce the carbon footprint of logistics operations. Environmental regulations are becoming stricter, moreover consumers are increasingly sensitive to the environmental impact of the products they purchase and how they are delivered. In response, many e-commerce companies are integrating sustainability into their logistics strategies. Sustainable logistics transportation practices are emerging as key solutions to reduce the environmental impact of freight movement in cities. It includes the use of electric or alternative-fuel vehicles, route optimisation through digital technologies, intermodal transportation systems and the development of urban distribution hubs.

These pressures on e-commerce companies to adopt sustainable last-mile practices do not emerge in isolation. They are shaped by a combination of regulatory requirements, competitive dynamics and evolving professional and societal norms. Understanding the institutional environment in which these adoption decisions are made is therefore essential to explaining not only whether companies adopt sustainable delivery, but why they do so and what forms that adoption takes (DiMaggio & Powell, 1983; Meyer & Rowan, 1977).

The adoption of such practices reflects a broader transition toward more sustainable logistics systems. However, implementing sustainable transportation solutions also affects how companies organise and manage their operations. Transportation decisions directly influence operational performance through delivery speed, logistics costs, service reliability and overall supply chain efficiency. For e-commerce companies operating in urban environments, deciding whether and how to transition toward environmentally sustainable delivery practices represents

a major strategic challenge. Understanding what drives these companies in this transition, what shapes their decisions, what determines their choice of solution and how they navigate implementation, remains insufficiently explored in academic literature.

1.2 Problem discussion

The rapid growth of e-commerce has significantly increased the volume of goods transported through last-mile delivery networks, particularly in urban areas. This expansion has improved delivery convenience for consumers but has simultaneously intensified the environmental impact of logistics operations. Cargo transport and especially road-based urban delivery, has become a major contributor to greenhouse gas emissions, increasing pressure on companies to adopt more sustainable practices (Van Loon et al., 2015).

E-commerce companies represent a particularly relevant focus for this study because, unlike traditional retailers, their business model is dependent on the physical delivery of goods to end consumers. Every order placed online generates a delivery trip, making last-mile logistics not a peripheral support function but a core operational activity. The environmental consequences of delivery decisions are therefore directly and immediately felt at the firm level, making e-commerce companies uniquely positioned to shed light on the challenges of sustainable last-mile adoption.

The urban context adds a further layer of complexity. Dense metropolitan environments are characterised by traffic congestion, restricted access zones, stringent local emission regulations and a higher delivery density. These conditions simultaneously intensify the environmental impact of conventional delivery and constrain the operational conditions under which sustainable alternatives can be deployed. It is precisely in these contexts that the pressure to transition is strongest and where the organisational challenges are most serious (Savelsbergh & Van Woensel, 2016).

Within this context, last-mile delivery concentrates the largest environmental footprint, the highest cost per parcel and the most significant operational complexity of the entire logistics chain (Savelsbergh & Van Woensel, 2016). It is therefore the stage at which sustainable adoption decisions carry the most weight and generate tangible organisational consequences.

Despite the growing attention regarding sustainable last-mile logistics, the existing literature remains predominantly technical and quantitative in nature. Indeed, it focuses on route

optimisation models, carbon emission calculations, simulation studies and vehicle performance comparisons. What remains largely absent is the organisational perspective, what institutional pressures shape how e-commerce companies experience, navigate and make sense of adopting sustainable last-mile delivery solutions in urban contexts (Tetteh et al., 2024; Thoma & Gruber, 2020). This gap justifies a qualitative inquiry into what institutional pressures shape urban e-commerce companies' approach to this transition.

1.3 Research question

What institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies in urban contexts?

1.4 Purpose

The purpose of this thesis is to explore what institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies in urban contexts.

This research focuses on last-mile delivery operations in metropolitan environments, where the pressure to adopt sustainable alternatives is most acute and the operational conditions are most complex. It is delimited to e-commerce companies that exercise meaningful control over their last-mile delivery strategy and have already initiated the process of implementing at least one sustainable delivery solution. This delimitation ensures that findings are grounded in direct organisational experience rather than hypothetical intent and allows the study to produce insights that are practically relevant and empirically grounded.

By examining what institutional pressures: coercitive, mimetic and normative are shaping the adoption decisions of e-commerce companies in urban contexts, this research aims to contribute to the academic literature on sustainable urban logistics while providing practical guidance for organisations navigating this transition.

1.5 Perspective

This research adopts an organisational perspective. The phenomenon of sustainable last-mile adoption is examined from the point of view of the companies and the professionals who lead

and experience this transition, rather than from the perspective of consumers, policymakers or logistics service providers.

1.6 Delimitations

This study is delimited along three dimensions. First, it focuses exclusively on e-commerce companies: firms whose logistics operations involve the physical delivery of goods. Second, the study is limited to companies operating in urban contexts. Rural or suburban logistics contexts are not addressed. Third, only companies that have already adopted at least one sustainable last-mile delivery solution are included, as the focus of the research is on the adoption process as experienced in practice rather than as anticipated or hypothetical.

No restriction is placed on company size, country of operation within Europe, or type of sustainable solution adopted. This deliberate diversity is consistent with the multiple case study design and is intended to capture a range of institutional contexts in which adoption decisions unfold, enabling cross-case comparison rather than within-case depth on a single organisational configuration.

1.7 Key terms

The following key terms define the conceptual boundaries of this research:

Sustainable last-mile delivery solutions: Transport and delivery approaches applied in the final stage of the supply chain, from distribution centre to end consumer, that aim to reduce environmental impact. In this thesis, these include solutions such as electric vehicles (EV), cargo bikes, urban consolidation hubs, relay points and grouped delivery models.

E-commerce companies: Firms whose primary sales channel is digital and whose logistics operations involve the delivery of physical goods.

Urban context: Dense metropolitan environments characterised by traffic congestion, limited delivery access windows and local emission regulations, which create specific operational constraints for last-mile logistics.

Adoption: The organisational process through which a company decides to implement, integrate and operationalise a sustainable last-mile delivery solution. In this thesis, adoption

encompasses the motivations behind the decision, the conditions shaping the choice of solution and the experience of implementation in practice.

2. Theoretical framework

This part reviews the relevant literature on sustainable solutions adoption drivers, barriers and conditions. It explores the concept of sustainable transportation in last-mile logistics and the e-commerce company as a decision-maker.

This chapter provides a comprehensive synthesis of existing academic research on the adoption of sustainable last-mile delivery solutions by e-commerce companies operating in urban contexts. As e-commerce continues to expand globally, the last-mile delivery stage has emerged as one of the most environmentally intensive and strategically significant segments of the urban logistics chain. This review examines how organisations navigate the decision to adopt sustainable delivery alternatives, including electric vehicles, cargo bikes, relay points and consolidated delivery models. Furthermore, it examines what factors shape these choices in practice. By mapping the current state of knowledge, this chapter identifies the drivers and barriers of sustainable last-mile adoption, contextualises the specific challenges of urban environments and highlights the research gap that motivates this study.

2.1 Literature review

2.1.1 Method of the literature review

To establish a rigorous and replicable theoretical foundation, a Systematic Literature Review (SLR) was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. This structured approach ensures transparency and academic reliability.

The identification phase was conducted through the Scopus database, selected for its extensive indexing of high-impact, peer-reviewed journals in the fields of Business Management and Supply Chain Logistics.

The search was executed using the following Boolean string:

“sustainab*” AND “last-mile delivery” AND (“adoption” OR “implementation”)

This string was designed to capture studies specifically addressing the adoption of sustainable solutions in last-mile delivery contexts, in direct alignment with the research question. This search strategy yielded an initial pool of 187 articles. These results were then screened in two stages.

In the first stage, titles and abstracts were reviewed to assess alignment with the research scope. Articles were retained if they addressed at least one of the following dimensions: the adoption of sustainable last-mile solutions by organisations, the drivers or barriers influencing such adoption, the implementation of sustainable delivery practices in urban contexts, or the organisational and strategic dimensions of sustainable last-mile logistics. Studies focused exclusively on technical optimisation, algorithmic modelling, drone technology, consumer acceptance or unrelated sectors such as agriculture, healthcare, or rural logistics were excluded at this stage.

In the second stage, full-text screening was applied to the remaining articles to verify depth of relevance and academic rigour. This two-stage process yielded 19 articles from the primary search. These were complemented by 11 articles identified through a manual search targeting ABS-ranked sources on four under-represented themes: organisational barriers to green e-commerce distribution, organisational framing of last-mile sustainability, institutional pressures in sustainable supply chains, and parcel locker and relay point adoption. This yielded a final corpus of 30 articles for full analysis.

Following selection, the 30 articles were analysed through a structured thematic reading process. Each article was summarised and annotated according to four analytical dimensions: the type of sustainable solution studied, the drivers and barriers identified, the urban context addressed and the organisational or managerial implications reported. Articles were then grouped thematically to identify convergences and divergences across the literature. Where multiple articles addressed the same phenomenon, their findings were compared and synthesised rather than reported individually.

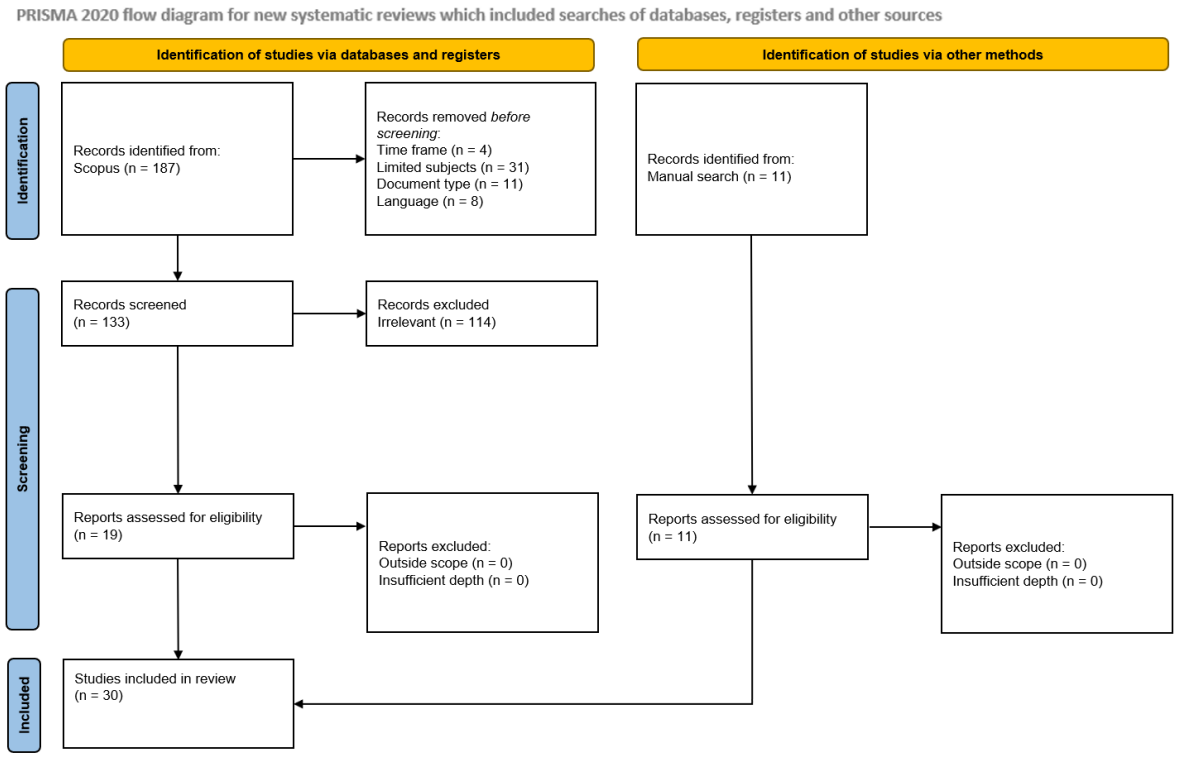


Figure 1: PRISMA Table

2.1.2 Thematic review of sustainable last-mile delivery adoption

The adoption of sustainable last-mile delivery solutions by e-commerce companies has become an increasingly central theme in logistics research. Urban environments tighten environmental regulations and consumers continue to demand rapid, reliable delivery. As a result, firms face increasing pressure to transition away from conventional road-based delivery models toward greener alternatives. This thematic review analyses the current literature across three dimensions directly relevant to the research question: the drivers that motivate the adoption, the barriers and challenges that constrain it and the conditions under which specific sustainable solutions are selected and implemented in practice.

2.1.2.1 Drivers of sustainable last-mile adoption

The literature identifies a range of factors that motivate e-commerce companies to adopt sustainable last-mile delivery solutions. These drivers operate at multiple levels: regulatory, economic, strategic and social.

Regulatory pressure is consistently identified as one of the most powerful external drivers. Indeed, the operational landscape in which e-commerce companies must function is being

reshaped by urban access restrictions, low-emission zones and increasingly strict national and local environmental legislation (Paddeu, 2025; Chatziioannou et al., 2024). Companies operating in cities such as Paris, Amsterdam or Stockholm face direct regulatory incentives and in some cases obligations to shift toward zero-emission delivery modes. Paddeu (2025) further shows that national decarbonisation strategies frequently misalign with local urban operational realities, meaning companies must actively navigate this gap. Valentini et al. (2026) demonstrate through an agent-based model that regulatory signals are among the most effective accelerators of eco-innovation diffusion in retail last-mile networks. It is particularly effective when combined with competitors' success in adopting sustainable solutions.

Economic motivations also play a significant role, though their influence is more ambiguous. While the upfront costs of sustainable solutions such as electric vehicle fleets remain high, several studies point to long-term cost advantages. Gao et al. (2022) demonstrate that incentivising EV adoption in city logistics requires alignment between pricing mechanisms, driving range policies and operational time window constraints, and that misalignment between these factors reduces adoption. Zero-emission vehicle adoption is increasingly driven by reduced fuel and operational costs over time, particularly as urban fuel taxes and congestion charges make conventional delivery more expensive. Thoma & Gruber (2020), in their qualitative study of cargo cycle adoption by German organisations, identify cost efficiency as a primary motivating factor alongside the ability to access restricted urban zones unavailable to motorised vehicles.

Strategic and reputational drivers are also prominent in the literature. Sustainability commitments have become a source of competitive differentiation. Thus, e-commerce companies increasingly use green delivery credentials to strengthen brand image and respond to environmentally conscious consumer expectations (Patella et al., 2021; Tetteh et al., 2024). Sharma et al. (2023) demonstrate that customer demand, corporate sustainability goals and regulatory pressure form three interconnected factors that collectively motivate last-mile modernisation.

Finally, the development of urban infrastructure includes the expansion of dedicated cycle lanes alongside the creation of urban consolidation centres with the deployment of low-emission zone exemptions for cargo bikes and EVs. These infrastructures are identified as important enabling conditions that are transforming sustainability intentions into operational decisions (Chatziioannou et al., 2024; Marujo et al., 2018).

2.1.2.2 Sustainable delivery solutions

The third theme concerns the specific solutions that e-commerce companies adopt and the organisational conditions that determine whether implementation is successful. The literature documents a range of sustainable last-mile alternatives, each carrying specific operational implications, adoption requirements and contextual dependencies.

Electric vehicles represent the most widely studied sustainable delivery solution. Rose et al. (2025) confirm that EVs are increasingly positioned as the default pathway toward decarbonisation in urban logistics, driven by regulatory incentives, improving battery technology and growing normative consumer expectations around sustainability. Marculetiu et al. (2023), in a systematic review of institutional pressures driving sustainable supply chain practices, further demonstrate that successful EV integration requires coordinated organisational decisions spanning fleet management, route planning and charging infrastructure. Highlighting that adoption is not only a procurement decision, but a systemic operational transformation shaped by strong institutional pressures.

Cargo bikes have emerged as a particularly relevant solution for dense urban environments. Iwan et al. (2021) demonstrate a 30% operational efficiency gain for cargo bikes over conventional vans in city centres. This can be linked to their ability to use cycling infrastructure and navigate restricted zones. Castrellon and Sanchez-Diaz (2024) further confirm that urban freight curbside management policies, including low-emission zones and delivery bay restrictions, create differentiated coercive pressure reshaping last-mile operations, while Gil et al. (2023) document a real-world network expansion case in Lisbon. Both illustrate how data-driven decisions and urban density patterns shape the strategic rollout of cargo bike operations. Garus et al. (2024) provide a comparative perspective, showing that while cargo bikes offer the lowest carbon footprint per parcel in dense urban contexts, their speed and load limitations mean that their adoption is most effective when combined with urban consolidation hubs.

Crowdshipping represents a more novel adoption pathway. Simoni et al. (2020) evaluate its potential as a sustainable last-mile alternative, concluding that its viability is highly dependent on urban density and participation rates. It makes it a solution that depends on the situation rather than one that works everywhere.

Relay points and parcel lockers represent a distinct and growing category of sustainable last-mile solution. By consolidating deliveries at fixed collection points, they reduce failed delivery attempts, shorten vehicle routes, and significantly lower the carbon footprint per parcel compared to home delivery models (Iwan et al., 2021; Ma et al., 2024). While most e-commerce companies offer relay points as an optional consumer choice, a subset have adopted mandatory relay-only delivery models as a deliberate firm-level organisational decision, embedding sustainability into their business model rather than leaving it to consumer discretion. Rose et al., 2025 show that such company-imposed models generate distinct normative institutional dynamics, with adoption driven primarily by internal sustainability commitments. Rizzitello et al. (2025) further demonstrate that mandatory sustainable delivery options yield stronger adoption outcomes than optional models, while Qi et al. (2025) confirm that company-driven mode selection significantly improves environmental performance relative to consumer-driven selection.

Across all solution types, the literature converges on a key finding: the success of sustainable last-mile adoption is not determined by the technical characteristics of the solution alone, but by the degree to which it is embedded in a coherent organisational strategy that accounts for urban context, operational constraints and organisational approach (Thoma & Gruber, 2020; Sharma et al., 2023; Patella et al., 2021).

2.1.2.3 Barriers and challenges of adoption

Despite the growing progress behind sustainable last-mile delivery, the transition remains constrained by a complex set of barriers covering operational, financial and infrastructural dimensions.

The barriers to electric vehicle adoption are extensively documented in the literature. Dadashzada et al. (2025) identify different barriers: from range anxiety to charging infrastructure gaps and higher upfront investment costs, these are the primary organisational obstacles preventing e-commerce and logistics companies from switching to full EV fleets. These findings are reinforced by Anosike et al. (2023), who demonstrate that charging downtime directly conflicts with the operational mandate of delivery reliability, which is a particularly strong tension in urban environments where delivery windows are tight and failed deliveries are costly. Alsaleh (2025) further highlights that the spatial organisation of urban

areas, including limited access to charging infrastructure within city centres, creates additional implementation complexity that companies must navigate when planning EV transitions.

Cargo bike adoption faces a distinct set of barriers. Thoma and Gruber (2020) identify few of those: load capacity limitations, weather dependency and the need of driver training as recurring operational constraints. These findings are shared by Michalakopoulou et al. (2025), who argue that cargo bikes require deliberate operational restructuring including the redesign of hubs, before they can function as viable substitutes for conventional vans in last-mile operations.

Beyond solution-specific barriers, the literature points to broader organisational and institutional challenges. Sallnäs and Björklund (2023), in a multiple case study of retailers, identify six interdependent barrier categories: organisational, financial, market, consumer, governmental and technological, preventing companies from offering green distribution alternatives, showing that no single intervention resolves the adoption challenge. Meyer-Waarden et al. (2026) further demonstrate that internal readiness, including managerial commitment, available resources and organisational culture, is as decisive as external pressure in determining whether a company successfully transitions to sustainable logistics, and that normative pressures often precede coercive ones in driving adoption.

The time-sustainability tension identified across the literature further intensifies these barriers. Consumer expectations for speed continue to constrain the operational space available for sustainable adoption. Indeed, consolidated or slower delivery models are typically more carbon-efficient but may conflict with service level commitments (Rose et al., 2025). This tension is particularly pronounced in e-commerce, where next-day or same-day delivery has become a competitive baseline in many urban markets.

2.1.2.4 The E-commerce company as a last-mile decision-maker

Within the broader landscape of urban logistics, e-commerce companies occupy a specific and underexplored position as last-mile decision-makers. Unlike traditional retailers or third-party logistics providers, e-commerce firms are directly responsible for the customer experience associated with delivery. Yet, they frequently rely on external carriers to execute it. This creates a particular decision-making dynamic in which firms must choose between internalising

sustainable delivery capabilities, such as operating their own cargo bike fleet or externalising them through partnerships with sustainable logistics providers (Heikkinen, 2024).

This distinction matters for research because the nature of the adoption decision varies significantly depending on the organisational model. On the one hand, companies that internalise sustainable solutions face operational transformation challenges. On the other hand, those that externalise must navigate through their partners selection and the service level agreements alongside the alignment with their sustainability values. In both cases, however, the strategic intent and organisational approach to adoption remain firm-level phenomena that qualitative inquiry is best positioned to explore (Thoma & Gruber, 2020; Sharma et al., 2023).

2.1.3 Research gap

The existing literature on sustainable last-mile logistics has grown substantially, yet it remains predominantly quantitative and technical in orientation. The majority of studies focus on optimisation models, carbon emission comparisons, simulation studies and vehicle performance analyses. These contributions are valuable but leave a gap in the organisational understanding of how adoption actually takes place within organisations (Tetteh et al., 2024; Patella et al., 2021).

Specifically, there is a lack of qualitative research examining how people in urban e-commerce companies experience the process of adopting sustainable last-mile delivery solutions. What motivates their decisions? How do they navigate the associated challenges? What shapes their choice of solution? While Thoma & Gruber (2020) provide a qualitative perspective on cargo cycle adoption by German organisations and Sharma et al. (2023) offer a qualitative analysis of drivers in last-mile modernisation, neither study focuses specifically on e-commerce companies in urban contexts. This thesis responds to that gap by adopting a qualitative, interview-based approach to examine how urban e-commerce companies approach the adoption of sustainable last-mile delivery solutions in practice and what institutional pressures shape those adoptions.

The review of the existing literature suggests that sustainable last-mile adoption cannot be fully explained by the technical or economic characteristics of the solutions themselves. The drivers identified in the literature: regulatory pressure, competitive imitation, and internal values map directly to the three mechanisms of institutional isomorphism described by DiMaggio and

Powell (1983): coercive, mimetic, and normative. The barriers identified, from the organisational resistance to the cultural readiness suggest that adoption is shaped by the institutional environment in which firms operate rather than by rational optimisation alone. Institutional Theory is therefore selected as the analytical framework for this study because it provides a systematic and theoretically grounded vocabulary for analysing precisely the types of external and internal pressures that the literature identifies as central to sustainable last-mile adoption. This theoretical choice directly informs the research question: by asking what institutional pressures shape e-commerce companies' approach to the adoption of sustainable last-mile delivery solutions, this study seeks to uncover which institutional pressures dominate, how they interact, and what this reveal about the conditions under which sustainable last-mile transition occurs in practice.

2.2 Institutional Theory

The theoretical framework of this research is grounded in Institutional Theory, as originally developed by DiMaggio and Powell (1983). This framework offers the analytical lens through which this study examines what institutional pressures shape the adoption of sustainable last-mile delivery solutions in urban contexts.

Institutional Theory emerged as a reaction to purely rationalist accounts of organisational behaviour, which assumed that firms make strategic decisions based solely on efficiency calculations and internal capabilities. Meyer and Rowan (1977) showed that organisations adopt formal structures not because they are necessarily efficient but because they confer legitimacy in the eyes of external actors. Based on this insight, DiMaggio and Powell (1983) extended it to explain why organisations within the same sector tend to become increasingly similar over time. They introduced the concept of the organisational field, which captures the collective set of actors operating within a common institutional space; including producers, customers, regulators and adjacent organisations that share recognisable norms and interact regularly. Organisations within a field are subject to similar institutional pressures and tend to converge toward homogeneous structures and practices through a process the authors term isomorphism.

Institutional Theory posits that organisations do not make strategic decisions in isolation. Rather, they operate within broader institutional environments composed of regulatory

systems, industry norms and social expectations that collectively shape organisational behaviour. A key driver of isomorphic change is organisational uncertainty: when organisations are unclear about the best course of action as is often the case with emerging practices such as sustainable last-mile delivery, where no single dominant solution has yet been established, they look to their institutional environment for guidance. This uncertainty is particularly acute in the context of urban last-mile logistics, where regulatory frameworks are evolving rapidly, competitive norms around sustainability are still being established and the technical and economic performance of alternative delivery solutions remains contested (Valentini et al., 2026; Tetteh et al., 2024). Under these conditions, institutional pressures play a central role in shaping how organisations approach adoption decisions. Facing such uncertainty, organisations tend to conform to the practices of their institutional environment through a process known as isomorphism, which operates through three distinct mechanisms (DiMaggio & Powell, 1983).

Coercive isomorphism refers to pressures exerted by external authorities: governments, regulatory bodies and urban administrations. DiMaggio and Powell (1983) define coercive pressures as those originating from both regulatory mandates and informal expectations imposed by entities on which the organisation depends, such as governmental authorities, parent corporations or major commercial partners. Paddeu (2025) demonstrate that national and urban access management regulations directly influence logistics operators' vehicle adoption decisions, though their effect is moderated by local operational conditions, suggesting that coercive isomorphism is a significant but contextually contingent driver of sustainable last-mile adoption. Importantly, the coercive mechanism does not require that all firms be directly subject to a regulation: the anticipation of future regulatory change, or the observation that competitors are adapting to regulatory pressure, can itself generate isomorphic behaviour (DiMaggio & Powell, 1983).

Mimetic isomorphism refers to the tendency of organisations to imitate the practices of peers and competitors when facing uncertainty. This mechanism is most active when organisations are uncertain about the best course of action and when clearly successful models are visible within the organisational field. DiMaggio and Powell (1983) note that mimetic processes often occur through indirect channels: the mobility of employees, the activities of consulting firms or simply the observation of what respected organisations are doing. In the last-mile logistics field, the large-scale adoption of electric vehicle fleets by Amazon creates a visible model that smaller e-commerce operators may feel pressure to emulate. Valentini et al. (2026) confirm

that the success of early adopters accelerates eco-innovation diffusion across retail last-mile networks. However, mimetic isomorphism does not necessarily mean passive imitation: organisations may selectively adopt elements of a practice while adapting others to their specific operational context, resulting in partial or modified forms of isomorphism.

Normative isomorphism refers to pressures arising from shared professional norms, consumer expectations and internal organisational values that push firms toward environmentally responsible practices. DiMaggio and Powell (1983) link this mechanism primarily to professionalisation: the process by which members of a profession build shared standards of practice, develop a body of knowledge and assert their authority over their occupational domain. In the context of sustainable logistics, normative pressures operate through multiple channels: the growing consensus within the logistics and e-commerce industry that sustainability is a strategic and reputational imperative; the expectations of consumers who increasingly evaluate the environmental credentials of their delivery providers; and the internal values and founding commitments of organisations whose leadership has made sustainability a core element of their identity. Tetteh et al. (2024) show that normative pressures, including customer expectations and corporate sustainability goals, are among the most consistent predictors of sustainable logistics adoption across different firm types and contexts. This is particularly relevant for the present study, as preliminary empirical evidence suggests that normative pressures may precede both coercive and mimetic mechanisms in driving sustainable last-mile adoption.

In this framework, the adoption of sustainable last-mile delivery solutions is understood not as a purely rational optimisation decision, but as an institutionally embedded response to a complex set of external and internal pressures. The unit of analysis is the organisation, specifically, the company as an actor within the urban logistics field rather than the individual professional. Interviews with individual participants serve as a means of accessing organisational-level data: respondents are selected for their direct knowledge of and involvement in last-mile delivery decisions, and their accounts are treated as windows into organisational experience and perspective rather than as personal opinions. This analytical stance is consistent with the organisational-level theory of isomorphism proposed by DiMaggio and Powell (1983), who explicitly situate their analysis at the level of the organisational field rather than the individual actor. This perspective directly guides the formulation of interview questions, the collection of empirical data and the analysis of findings, allowing this research to examine what institutional pressures shape the adoption of sustainable last-mile delivery

solutions by e-commerce companies in urban contexts, and how these pressures interact across different organisational configurations.

3. Methodology

This section describes the research philosophy, approach and methods used in the study. It details the qualitative research design, data collection through semi-structured interviews and the thematic analysis employed to interpret the findings.

3.1 Research philosophy

This research adopts an interpretivist philosophical stance. Ontologically, interpretivism rejects the view of reality as a single objective truth and instead understands it as constructed through individuals' experiences, perceptions and meaning-making (Saunders et al., 2019). Epistemologically, this implies that knowledge is generated through close engagement with research participants rather than through objective measurement. The way e-commerce companies approach the adoption of sustainable last-mile delivery solutions is experienced differently depending on firm size, urban context, institutional pressures and organisational priorities and can therefore only be understood through direct engagement with the professionals who navigate these decisions. This philosophical orientation justifies the qualitative methodology adopted in this study and shapes all subsequent methodological choices, from data collection to analysis.

3.2 Research approach and design

This study follows a qualitative research design. Qualitative research is concerned with understanding meaning, experience and context rather than measuring variables or establishing statistical relationships (Bryman, 2016). Given that this thesis seeks to understand what institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies, from the perspective of the professionals who lead and experience these transitions, a qualitative approach is the most coherent and rigorous choice available.

In terms of reasoning, this study adopts an abductive approach. Abductive reasoning combines theoretical sensitivity with openness to empirical discovery (Saunders et al., 2019). This study

begins with an established theoretical lens, Institutional Theory, which orients data collection and analysis, while remaining open to findings that may enrich, nuance or challenge the framework. This is particularly appropriate here, as the three mechanisms of Institutional Theory serve as sensitising concepts rather than rigid predetermined categories: they guide the inquiry without preventing the possibility of unexpected insights emerging from the data.

More specifically, this study adopts a multiple case study design (Yin, 2018), in which each interviewed organisation constitutes an individual case. Yin (2018) argues that the case study approach is particularly appropriate when the research investigates contemporary phenomena in their natural setting, and when the inquiry focuses on what, how or why questions that cannot be separated from their context. A multiple case design was selected because it enables cross-case comparison, allowing the study to identify both convergent patterns and meaningful differences in how different e-commerce companies approach sustainable last-mile adoption. This comparative logic is directly consistent with the analytical objectives of Institutional Theory, which predicts that organisations within the same institutional field will tend toward isomorphism while potentially responding to pressures differently depending on their size, values and context.

3.3 Data collection

Primary data was collected through semi-structured interviews conducted with professionals and operational decision-makers in e-commerce companies that have already implemented sustainable last-mile delivery solutions. This method was selected as the most appropriate means of generating the in-depth data required to address the research question.

Semi-structured interviews offer a balance between consistency and flexibility (Saunders et al., 2019). A pre-established interview guide ensured that all participants were asked about the same core themes (*Appendix 1*). For instance, the motivations behind their sustainable delivery decisions, the conditions that shaped their choice of solution and their experience of implementation in urban contexts. Naturally, the conversation to further develop based on each respondent's specific experience and context were allowed (Saunders et al., 2019). This flexibility is valuable in an exploratory study, where unexpected insights from participants can be as analytically significant as anticipated ones.

Interviews were conducted via video conferences. Google Meet was used to ensure a stable and professional communication environment. Interviews were conducted in French, according to the respondent's preference. Conducting interviews in the participant's preferred language was considered essential to minimise misunderstandings and to allow respondents to articulate nuanced professional experiences as naturally and precisely as possible. All interview data collected in French was subsequently translated into English by the research team prior to analysis, with the assistance of artificial intelligence tools (Claude AI).

Each interview lasted approximately 25 to 45 minutes and was recorded with the explicit consent of the participants. Recordings were then fully transcribed to enable systematic and rigorous thematic analysis. A target of ten interviews was established, consistent with the standards of qualitative research at this level of study, where depth of insight takes precedence over sample size (Bryman, 2016). Data collection continued until theoretical saturation was approached. That is, until additional interviews ceased to generate substantially new themes or perspectives.

3.4 Sampling

The selection of participants followed a purposive sampling strategy, designed to identify respondents with direct experience and decision-making responsibility in sustainable last-mile logistics. Purposive sampling is appropriate for qualitative exploratory research where the objective is to generate depth of insight rather than statistical representativeness (Saunders et al., 2019).

The sampling process operated on two distinct levels, consistent with the multiple case study design described in Section 3.2. At the first level, organisations were selected based on organisational criteria. At the second level, individual participants were selected within each organisation based on role-specific criteria. Both levels required a clear rationale, as the validity of the data depends both on choosing the right organisations to study and on accessing the right people within those organisations (Yin, 2018).

At the company level, three criteria governed selection. First, the organisation had to be an e-commerce company whose sales channel is digital and whose operations involve the direct delivery of goods to end consumers. Second, it must have already adopted at least one

sustainable last-mile delivery solution. This criterion was essential to ensure that selected organisations could speak from direct operational experience rather than hypothetical intent. Companies that had expressed interest in sustainable delivery but had not yet implemented any solution were excluded. Third, the company had to exercise meaningful organisational control over its last-mile delivery strategy. Companies that impose a mandatory relay-point or parcel locker model as their sole delivery option, making this a firm-level organisational decision rather than a consumer choice are included in the scope, as this constitutes a deliberate sustainable adoption strategy. Conversely, companies relying exclusively on customer-initiated sustainable options such as relay points selected at checkout by the end consumer were excluded, as sustainable outcomes in such cases result from consumer choice rather than a firm-level organisational decision. In addition, the sample was designed to capture diversity across company size, geographic location and type of solution adopted. This deliberate variation is consistent with the theoretical logic of Institutional Theory, which predicts that organisations operating in different institutional sub-environments will be exposed to different combinations of coercive, mimetic and normative pressures. Capturing this variation across cases enhances the analytical richness of the cross-case comparison (Yin, 2018).

At the individual level, participants were required to hold a position with direct involvement in or meaningful proximity to last-mile delivery decisions. Rather than applying a strict seniority criterion, the selection prioritised sufficient operational knowledge of the adoption process, whether that knowledge derived from strategic responsibility, day-to-day operational involvement or close cross-functional exposure to delivery decisions. This approach reflects the heterogeneity of the sample: in a five-person SME, a general coordinator may hold as much practical insight into adoption decisions as a dedicated logistics director in a large multinational. Participants were also required to have been directly involved in, or closely informed about, the company's adoption process, to ensure accounts were grounded in lived organisational experience rather than retrospective observation.

The purposive sampling strategy carries inherent limitations that must be acknowledged. At the company level, the sample is not statistically representative of the broader e-commerce population, findings should not be interpreted as prevalence estimates. Furthermore, organisations were identified primarily through professional networks and LinkedIn outreach, which may introduce a self-selection bias: companies with a stronger sustainability orientation

or more advanced adoption processes may be more likely to accept interview requests, potentially overrepresenting adopters in the sample.

Reference	Company	Date of the interview	Platform used	Duration (in min)
Participant 1	Company 1	28/03/2026	Google Meet	**38**
Participant 2	Company 2	07/04/2026	Google Meet	**23**
Participant 3	Company 3	15/04/2026	Google Meet	**41**
Participant 4	Company 4	15/04/2026	Google Meet	**28**
Participant 5	Company 5	19/04/2026	Google Meet	**28**
Participant 6	Company 6	20/04/2026	Google Meet	**24**
Participant 7	Company 7	20/04/2026	Google Meet	**30**
Participant 8	Company 8	15/04/2026	Google Meet	**32**
Participant 9	Company 9	21/04/2026	Google Meet	**25**
Participant 10	Company 10	22/04/2026	Google Meet	**39**
Participant 11	Company 11	23/04/2026	Written responses via e-mail	**n/a**

Figure 2: Sample selection for interview

3.5 Data analysis

The data collected through semi-structured interviews was analysed using thematic analysis, following the six-phase framework proposed by Braun and Clarke (2006). This method was selected for its flexibility and suitability for exploratory qualitative research, as it enables patterns of meaning to emerge from the data through an inductive process of reading, coding and grouping, rather than imposing predetermined categories onto the material.

The analysis began with a data-driven phase. In the first phase, all interview transcripts were read line by line, in full, multiple times to achieve deep familiarity with the data and develop an initial sense of the patterns present across respondents. In the second phase, initial codes were generated systematically by identifying meaningful segments of text across the entire dataset. Each transcript was coded openly, allowing first-order codes to emerge directly from the participants' language and experiences. In the third phase, first-order codes were grouped into higher-level categories, referred to as second-order codes, reflecting recurring patterns across respondents. In the fourth phase, these second-order codes were reviewed and refined to ensure internal coherence and clear distinction from one another. In the fifth phase, second-order codes were organised into final themes, defined and named with precision. In the sixth and final phase, the analysis was written up and connected to the institutional theory. The full

coding structure, documenting the progression from first-order codes to second-order codes to final themes, is presented in Appendix 2.

While the initial coding was data-driven, the analytical process was subsequently guided by the three mechanisms of Institutional Theory: coercive, mimetic and normative isomorphism. They oriented the organisation of final themes without preventing new or unexpected patterns from emerging from the data. For instance, if a respondent described pressures not clearly captured by any of the three mechanisms, these observations were coded and retained as potentially enriching the analysis rather than being discarded.

The thematic analysis was structured across two distinct analytical levels, consistent with the multiple case study design described in Section 3.2.

At the first level, a within-case analysis was conducted for each of the eleven cases individually. Each transcript was examined as a standalone unit, allowing the specificities of each organisation: its size, sector, urban context and adoption trajectory, to be captured. This within-case layer forms the basis of Chapter 4, where empirical data is presented thematically but grounded in case-specific evidence, setting the scene for the cross-case analysis that follows.

At the second level, a cross-case analysis was conducted to identify patterns, convergences and divergences across the eleven cases. Once within-case themes had been established, the research team systematically compared how each theme manifested across different companies, looking for shared configurations as well as notable exceptions. By combining within-case depth with cross-case breadth, the analytical approach ensures both the faithful representation of individual experiences and the generative potential of pattern recognition across a diverse sample (Yin, 2018).

3.6 Research trustworthiness

Following Lincoln and Guba (1985), trustworthiness is assessed across four dimensions: credibility, transferability, dependability and confirmability.

Credibility refers to the accuracy with which the findings reflect the perspectives of the participants. To enhance credibility, interviews were conducted in the respondent's preferred language to minimise misunderstandings. Moreover, participants were encouraged to speak freely and elaborate on their experiences. The use of multiple respondents across diverse organisations further strengthens credibility by providing a range of perspectives on the phenomenon under study.

Transferability refers to the extent to which findings can be applied to other contexts. While qualitative research does not aim for statistical generalisation, transferability is enhanced through description, the provision of detailed contextual information about the research setting, participants and findings. Allowing readers to assess the relevance of the study to their own contexts (Bryman, 2016). This thesis provides detailed descriptions of the organisational and urban contexts in which the studied companies operate.

Dependability refers to the consistency and transparency of the research process. To address this, a clear and documented methodology has been established throughout this thesis. It includes explicit criteria for participant selection detailed in Section 3.4, a structured and bilingual interview guide provided in Appendix 1 and a systematic six-phase thematic analysis process outlined in Section 3.5. Furthermore, all interviews are recorded with participant consent and fully transcribed prior to analysis, ensuring that the analytical decisions made throughout the study are grounded in a stable and consistent empirical foundation.

Confirmability refers to the degree to which findings reflect the data rather than the researchers' own biases. To enhance confirmability, the coding process was conducted independently by each member of the research team before being compared collectively. Then, all analytical decisions were discussed and justified. The use of Institutional Theory as a theoretical framework further anchors the analysis in an established scholarly tradition rather than subjective interpretation.

3.7 Ethical considerations

Ethical integrity is a fundamental requirement of academic research and was maintained throughout all stages of this study, from the design of the interview guide to the storage and

reporting of data. The following section outlines the key ethical principles that guided the research process.

All participants were contacted in advance and provided with a clear explanation of the study's purpose, scope and intended use of the data before agreeing to participate. Participation was entirely voluntary. Prior to each interview, participants were explicitly informed that they could decline to answer any question, request that specific statements not be included in the analysis, or withdraw from the study at any time. Written or verbal consent was obtained before recording began. For the participant who responded in writing (P11), consent was implied through the act of responding and confirmed in the introductory exchange. In addition, GDPR requirements were collected for all participants.

Given that participants shared professional information about internal logistics practices, supplier relationships, cost structures and strategic decisions, protecting their identity was both an ethical obligation and a prerequisite for obtaining responses. All participants and organisations were fully anonymised in the transcripts and throughout the thesis. Individuals and organisations are referred together as Case 1 to Case 11. Where a company name appears within a direct interview quote, it has been replaced with [Anonymised]. This level of anonymisation was adopted to ensure compliance with GDPR requirements and to protect both individual and organisational confidentiality.

In line with the principle of data minimisation, data collection was limited strictly to information relevant to the research question. Interview questions focused on professional practices, organisational decisions and perceived motivations related to sustainable last-mile delivery. No sensitive personal data including racial or ethnic origin, health, political opinions was solicited or recorded. Where participants voluntarily shared information beyond the scope of the research question, this information was not included in the analysis.

This study was conducted in accordance with the General Data Protection Regulation. Participants were informed of their rights under GDPR, including the right to access, correct or request deletion of their data. All recordings and transcripts were stored securely, accessible only to the three members of the research team and used exclusively for the purposes of this thesis. Data will not be retained beyond the completion of the academic assessment process. The data processing application required by Jönköping University was completed and signed by both the research team and the thesis supervisor and is included as an appendix to this thesis.

As qualitative researchers, the authors acknowledge that their own backgrounds, values and familiarity with the French e-commerce and logistics landscape may have influenced both the design of the study and the interpretation of the data. All three members of the research team have personal and professional connections to the French market, which facilitated access to participants but may also have introduced implicit assumptions about what constitutes a “normal” or “expected” approach to sustainable delivery. This reflexive awareness was maintained throughout the analytical process, particularly during the thematic coding stage, where interpretations were cross-checked among the three authors.

Interviews were conducted in a respectful and non-coercive manner. Questions were formulated to be open and non-leading, and participants were not pressured to express opinions aligned with the researchers' theoretical framework. Where participants expressed views that contrasted with the dominant patterns in the data these responses were recorded and reported faithfully rather than minimised.

4. Empirical Findings

This chapter presents the first level of analysis, in which the empirical data from eleven semi-structured interviews is organised thematically to set the scene for the cross-case analysis conducted in Chapter 5. Data is structured following the six-phase thematic analysis described in Section 3.5. The full thematic coding table documenting the progression from first-order codes to second-order codes to final themes is presented in Appendix 2.

4.1 Overview of cases

Before presenting the thematic findings, this section provides an overview of the eleven cases interviewed, their sustainable solutions and the participants who contributed to this study. This contextual summary allows the reader to situate each case and understand the diversity of the sample.

Case	Participant	Role	Company size	Sector	Urban context	Sustainable solution(s) adopted	Delivery model
Case 1	P1	Senior Analyst, Operation Finance	Large	General e-commerce	Europe	Electric vans, cargo bikes	In-house & Outsourced
Case 2	P2	Supply Chain & Logistics	SME	Organic food, zero-waste	Nationwide, (France)	Cargo bikes, electric vehicles	In-house
Case 3	P3	Transport coordinator	Large	Fashion logistics	Nationwide (France)	Green carriers, biofuel shuttles, cargo bikes	Outsourced
Case 4	P4	Manager (Commerce & Delivery)	Micro	Vintage crockery selling & rental	Lille, France	Electric van, electric car	In-house
Case 5	P5	Supply Chain & Transport	SME	Beauty / nail care	Paris, France	Cargo bikes, electric trucks	Outsourced
Case 6	P6	Founder & CEO	Micro	Second-hand children's books	Nationwide (France)	Relay-point model (Mondial Relay)	Outsourced
Case 7	P7	Procurement & Logistics	SME	Niche perfumery	Paris, France	Cargo bikes (Paris only)	Outsourced
Case 8	P8	Head of Transport France	Large	Eyewear / optics	Nationwide (France)	Electric vans, bike deliveries (select cities), night delivery, flow consolidation	Outsourced
Case 9	P9	Co-founder	Micro	Artisanal bone broth	Lille, France	Cargo bike	In-house
Case 10	P10	Sales Admin & Operations	SME	Premium artisan chocolate	Paris / Île-de-France	Consolidated delivery route (TDG)	Outsourced
Case 11	P11	Co-founder	Micro	Responsible fashion	Nationwide, (France)	Free delivery model (imposes sustainable carrier choice), decarbonised shipments	Outsourced

Figure 3: Case overview

The sample encompasses significant variation in company size, from micro-enterprises of fewer than five employees to large multinationals operating across continents. The sectors represented range from food and beauty to fashion, optics and logistics services. Geographically, most cases operate in French urban contexts. This diversity is consistent with the research design described in Section 3.2 and enables the cross-case comparison that the analysis in Chapter 5 will undertake.

4.2 The selection and type of sustainable solutions

The eleven cases adopted a range of sustainable last-mile delivery solutions. These can be grouped into four categories: electric vehicles, cargo bikes, relay-point and consolidation models, and strategic carrier selection.

4.2.1 Electric Vehicles

Electric vehicles were adopted by Case 1 (electric vans), Case 2 (own EVs), Case 4 (electric van and car), and to a limited extent by Case 8 (pilot projects in select cities) and Case 5. Case 1 stands out for scale: P1 stated that "*a zero-emission delivery means either micromobility, the cargo bikes in the city centre that allow us to reach areas that are probably more restricted, or the second option, which is electric vans, the [Electric van]*". He estimated that "*over 50% of deliveries*" in Munich were already zero-emission.

Case 2 adopted electric vehicles progressively as the company grew. P2 explained: "*as the company grew, our company got a lot of attention with the expansion and with more and more customers, we started purchasing electric vehicles. And today we are very happy about that, especially given the rising price of petrol*".

Case 8 was at an earlier stage: "*One of the projects we tried to carry out, which is still ongoing, is the use of electric trucks on certain flows. It is something quite recent, which has been implemented in Italy on a specific flow where there are many back-and-forth trips and large volumes*".

4.2.2 Cargo bikes

Cargo bikes were used by Case 1, Case 2, Case 3, Case 5, Case 7 and Case 9. For Case 7, cargo bikes were the exclusive solution in Paris: "*In urban areas we have preferred to use cargo bikes, to promote sustainable development and to reduce our carbon footprint*". P7 added that the solution originated during Covid: "*It was actually during Covid that we really reflected on ways to reduce our impact and to operate differently in terms of delivery*".

Case 3's cargo bike use operated through a contracted partner: "*In Saint-Malo, it's exclusively trikes for delivery. In Rennes too. [...] they come to collect parcels with small vans, sometimes electric and bring them to their urban micro-depots [...] from which they load the cargo bikes and do the last mile by bike*".

Case 1 also deployed cargo bikes in city centres to reach areas inaccessible to vans. P1 explained: "*we have micromobility, primarily the Cargo Bike, which allows us to go into city centres in areas that are for example pedestrian-only, or where bicycles can circulate but where vehicles are not necessarily welcome. This allows us to deliver directly to more customers*".

P2 began with cargo bikes as the primary solution before expanding to electric vehicles, stating: "*we are mainly in city centres, so we use cargo bikes as much as possible, especially in the earlier period*".

Case 9's entire delivery model is built around the cargo bike. P9 described the choice as self-evident: "*I simply could not imagine delivering any other way than by bike. In any case, my clients are within a 5 km radius as I explained. [...] the cargo bike was a solution, the question did not even arise. It was obvious*".

4.2.3 Relay-point and consolidation models

Case 6 adopted relay-point delivery as its primary model through Mondial Relay. P6 described a nudging strategy to achieve this: "*The principle is to include delivery in the price. [...] By including delivery in the box price, customers by default choose the relay-point option*". P6 made an important distinction in how this is communicated: "*By including delivery in the price, the delivery becomes 'offered' rather than free. And it is important to say it is offered rather than free, in terms of messaging, free makes delivery seem cheap, whereas offered makes it feel like an additional service provided by the company*". This framing reveals a deliberate behavioural mechanism embedded in the pricing strategy.

P10 developed a consolidated delivery system in Île-de-France: "*what we call the TDG, a large trailer belonging to our logistics provider [Anonymised]. It follows a fixed route within Île-de-France on specific collection and delivery days. This allows us to serve all our major Île-de-France accounts [...] in a consolidated way rather than making dozens of individual trips*".

Case 11's free-delivery model functioned as a structural mechanism: "*When delivery is not included in the price, customers tend to choose the least ecological delivery option because it is the cheapest. [...] By offering free delivery, we therefore impose a sustainable delivery method on customers regardless*".

4.2.4 Strategic carrier selection

For companies that outsourced delivery, the selection of sustainable carriers was the primary lever. P3 integrated sustainability into tender processes: "*When we run tenders on flows, this will be one of the key questions: how do you handle this topic*". He distinguished between credible and superficial claims: "*A carrier who tells us 'I deliver Paris and all the major cities by electric vehicle or low-emission vehicle' we go with them. If a carrier tells us 'To offset my CO2 emissions, I plant trees in the Amazon', I am a bit less keen*".

Case 8 described a similar approach: "*We regularly ask carriers what actions are being implemented. [...] It is our role within transport [...] to challenge carriers to obtain information on what is being implemented*".

P1 presented a unique case where the e-commerce company itself imposes the fleet on its delivery partners rather than selecting carriers based on sustainability: "*we have a defined set of vehicle types that we offer to our DSPs, Delivery Service Partners [...] So at certain Delivery Stations we only use [Electric van], and in that case we provide them to our delivery partners*". This reverses the typical outsourced model: the company functions as the source of coercive pressure toward its logistics partners, rather than delegating sustainability decisions to carriers.

4.3 Motivations behind the adoption of sustainable last-mile delivery

Across the eleven cases, participants described a variety of motivations for adopting sustainable last-mile delivery solutions. These motivations can be grouped into five recurring sub-themes: internal values and ecological awareness, customer expectations, competitive dynamics and peer influence, economic considerations and governmental regulation.

4.3.1 Internal values and ecological awareness

The most consistently cited motivation across the sample was a deep alignment between sustainable delivery and the company's founding values or identity. This was particularly pronounced among smaller companies whose business models were built around sustainability from inception.

At Case 2, P2 explained that sustainability was embedded in the company's DNA from the outset: "*[Anonymised] is a shop and a drive combined [...] The core idea is to offer food*

products in glass jars, to give customers that eco-responsible consciousness". When asked whether the transition to electric vehicles was regulation-driven, P2 was clear: "*No, it was not really regulation. [...] it was really customer demand. The company was created around a demand from customers to eat better*". P2 further added that reverting to thermal vehicles would cause tangible damage: "*Going back? Yes, that would definitely make us lose customers, people who no longer trust us and feel our values have changed. So yes, I think we would lose customers*". P2 also described a sense of organisational pride in living up to these values: "*it is genuinely a source of pride to represent the company and its values, and to see that the customer demand is met and that customers are happy and satisfied, both in terms of their food being local and genuinely a short, closed-loop circuit*".

Regarding Case 4, the alignment between delivery and values was equally strong. P4 stated: "*The main reason was above all that it is much more aligned with the fact that we work with second-hand items. The whole idea of not wasting, of circular economy*". When asked about regulatory influence, P4 was unequivocal: "*Even without any rules, we would have adopted the electric van anyway, because it is part of our company values*".

Case 9 presented the most value-driven case. P9 described the cargo bike choice as self-evident: "*I simply could not imagine delivering any other way than by bike*". P9 traced this conviction to personal history: "*I am from Eastern France and I did all my internships in Germany when I was young. The ecological awareness in Germany was very, very well developed*". The entire supply chain was designed for consistency: "*All the raw materials are sourced from farms around my home. [...] I had solar panels installed and I produce during the day to optimise production*". The coherence between product and delivery was, for P9, not a strategic choice but a logical necessity: "*It was not even a question. I could not imagine making a super virtuous broth and delivering it with an anti-virtuous solution. It simply did not fit the company's values at all*".

Case 11 adopted a distinctive structural mechanism to enforce sustainable delivery. P11 explained: "*From the very launch of the company, we made the deliberate choice to offer free delivery on every order, because this allows us to choose the delivery method ourselves. Having that choice on our side means we can select the most ecological delivery option available from the carrier, even if it costs us more*". P11 framed this commitment in the context of the broader fashion industry: "*We are aware that the fashion world is an ecological disaster and we do not*

want to add to it. Everything is thought through to reduce our carbon footprint as much as possible".

At Case 5, P5 framed sustainable delivery as a matter of brand coherence: "*It immediately comes down to aligning with the brand's values. [...] we position ourselves as green, as a brand that truly advocates responsible beauty. So obviously we cannot sell products and deliver them in diesel trucks*". However, P5 also acknowledged that the delivery side was still lagging behind product sustainability: "*there is still a real margin to make the whole brand green, truly sustainable. We are really more focused on the products than on the delivery, and there is still a lot of work to make that genuinely visible*". This honest assessment suggests that even within values-driven companies, the normative coherence between product and delivery is a work in progress rather than a fully achieved state.

Case 7 followed a similar logic. P7 described how the founders' personal practices directly shaped the delivery model: "*one of [the founders] has his own fields in Grasse. [...] Reflecting on how he operated in alignment with sustainable development, harvesting himself [...] the idea of implementing something in line with that vision and mindset came directly from the founders*".

Even at larger companies, values played a role, though with a different texture. At Case 1, P1 described the Climate Pledge as "*a genuine internal driver*" and connected it to a core management principle: "*Customer obsession currently means pursuing a greener delivery where possible*". P1 specified that sustainability was embedded in budgetary governance: "*Environmental questions are at the centre of the budget decisions we make*".

At Case 3, P3 described a cultural dimension: "*We are based near Saint-Malo, lots of sailors, people who love nature. So naturally there is an attraction to the topic*". P3 also revealed a long institutional trajectory: "*we have had someone responsible for CSR for a long time. At the beginning it was more global QHSE. But for the last 15 years, there has been a dedicated person. And the company is 20 years old*".

P8, despite being one of the largest companies in the sample, described the transition as voluntary rather than imposed: "*I would say it is chosen. We are not forced, these are decisions made by the company, even if they create challenges*". P8 also noted that sustainability was

reinforced through internal training: "*I would tend to say it is still a topic that comes up regularly. There are also training sessions on the subject. So we are sensitive to it*".

A recurring pattern across these accounts is the sense that sustainable delivery was not experienced as a strategic option but as an organisational necessity, a logical extension of the company's identity that could not be dissociated from the product or service itself. This pattern, and its analytical implications, will be examined further in Section 5.5.

4.3.2 Customer expectations

Customer expectations emerged as a significant but variable driver across cases. For companies whose customer base was already environmentally conscious, this factor reinforced and validated the internal values described above.

P2 explained: "*The company was created around a demand from customers to eat better*". P2 provided rare quantitative evidence of this demand: "*We ran a survey, a panel of about 1,000 customers, and we asked them, for example, if the price of a tomato were to increase, how much would they be willing to pay more? And honestly, I believe we had around 75% positive responses*".

P4 observed: "*There are clients who would not be particularly enthusiastic about delivery if it didn't align with their ethics or their image, if the crockery was new, bought from a wholesaler, or delivered by a big diesel truck*".

P5 considers heavily the customer expectations: "*We have a clientele that is relatively affluent in terms of what they buy, but also very interested in sustainable development and in really paying attention to what they put on their nails, and also what that generates downstream*".

P7 also pointed to the consumer's evolving view as a reinforcing factor: "*the consumer's gaze regarding our responsibility in this space. I think that helped a lot in putting this in place*". P7 also acknowledged that discontinuing sustainable delivery would have consequences: "*I think we would lose a certain segment of the population, on the principle of proposing something and then discontinuing it*". Yet P7 also highlighted a tension inherent to the luxury segment: "*we are also a company positioned in the luxury segment, and luxury is not the sector most aligned with sustainable development, or at least with implementing solutions that go with it*". This luxury-sustainability tension was further articulated: "*it can sometimes be difficult to*

manage from the consumer's perspective, they want something luxurious but also sustainable. And it is genuinely complicated to combine the two".

However, not all companies experienced strong customer demand for sustainable delivery. P8 offered a contrasting perspective: "*The customer today is not very demanding on this type of topic. What they mainly want is to receive their goods as quickly as possible and without damage*". P8 further observed: "*Today, green logistics is difficult mainly because customers expect speed. When you click, you want delivery the next day or immediately*".

P10 placed customer expectations last among the drivers discussed, explaining that clients buy "*because our overall brand values resonate with theirs. The way we deliver is part of that broader picture [...] not the primary selling point in isolation*".

P11 identified a structural paradox in customer behaviour: "*One problem we identified is that when delivery is not included in the price, customers tend to choose the least ecological delivery option because it is the cheapest*". This observation reveals that stated customer preferences for sustainability do not always translate into actual choices when price is a factor, a tension that directly motivated Case 11's free-delivery model.

This variation suggests that the strength of customer expectations as a driver depends on how directly the company communicates with its end consumers and on the degree to which the customer base is already sensitised to environmental concerns.

4.3.3 Competitive dynamics and peer influence

Across the sample, nearly all participants described sustainable delivery as a competitive advantage, even when competition was not a primary motivating factor for adoption.

P4 considered it a clear advantage: "*Coming back to our brand image and the perception people have when they come to us, it can be a real advantage. And no one is against ecology and reducing emissions*". P6 called it "*a differentiating and exclusionary criterion relative to the other offers on the market*". P7 agreed: "*It is really about standing out from competitors*".

P11 explicitly used sustainable delivery as a differentiator: "*When we were researching during the creation of our brand, we noticed that clothing brands do not communicate much about last-mile sustainability. [...] So yes, it influenced us, because we differentiate ourselves already*

through our delivery method alone". P11 went further: "*All the effort seems to go into recyclable and recycled packaging, but for us that was not enough*", positioning delivery itself as the overlooked frontier of sustainability in fashion.

P3 noted the value of formal certification: "*This allows the company to obtain labels like the EcoVadis label. We very recently renewed and obtained the Silver Medal*".

P9 described a different form of differentiation, rooted not in market positioning but in personal exemplarity: "*I think the solution is always influenced by a person through their own example and exemplarity. I think I touch people through the exemplarity of this way of offering an absolutely wonderful product in a hyper-ecological way*". P9 further observed that this approach was already generating indirect competitive effects in the local ecosystem: "*one of my clients recently bought a bike to cycle to work, because they realised they worked 5 km from home and that there might be an ecological reason to cycle*".

However, the origin of the competitive logic differed. For most SMEs and micro-enterprises, sustainability was adopted for internal reasons and then recognised as a competitive advantage afterwards. P4 illustrates this: the eco-responsible label "*was not really planned from the start, even if being able to get a label and have an additional argument to use [Anonymised] was an objective. But it was not the primary motivation, it was fundamentally based on our values*".

P2 described a first-mover logic: "*you should not wait. We were a little bit among the first on the market to do this. And the demand is genuinely there*". P2 framed this as an advantage of agility over scale: "*My main piece of advice is really to just go for it, not to keep waiting behind the big players who are ultimately more focused on performance and their budgets and much less on the real, concrete demand from customers*".

P1 presented an inverse logic. P1 positioned Case 1 not as a follower but as a pioneer: "*I would consider us more as pioneers, given the scale of our operations*". The competitive dynamic here was about maintaining leadership rather than catching up. P8 echoed this from a market leader perspective: "*Let's say we are the leaders in the sector, but that does not prevent us from taking inspiration from our competitors*".

While sustainability was widely perceived as a competitive advantage, the question of whether competitors directly influenced adoption decisions produced a different pattern. When asked

whether other companies in their sector influenced their adoption decisions, most participants reported limited direct peer influence. P4 stated: "*We don't really know any direct competitors*". P7 said: "*I am not sure. Personally, I do not know many similar cases*". P5 noted: "*That was a fairly minor aspect, because honestly we already had very strong drivers before we even had to worry about competitors*". At the same time, P3 acknowledged that the broader market was moving in the same direction: "*The big players on the market have green solutions too. It is becoming the norm*".

Case 6 stood out as the only case where competitive dynamics were explicitly mentioned as the primary driver. P6 stated: "*It is true that the ecological criterion, not just on delivery but more broadly on the choice of business model, was indeed somewhat motivated by a desire for differentiation relative to those competitors*". P6 placed "*competition, customers, and ecological awareness in first place, roughly in that order*". P6 further acknowledged that the ecological dimension would have mattered less a decade earlier: "*Taking into account that home delivery was cheaper back then, I probably would have prioritised it a bit less for economic reasons. And the ecological aspect, customers were also less concerned about it*". This temporal awareness suggests that P6's adoption decision was shaped by an evolving competitive and societal landscape rather than by values alone.

Case 10 provided an interesting case of indirect peer influence: "*We drew inspiration not from direct competitors but from wine distributors who share our logistics provider [Anonymised]. They face the same temperature and handling challenges we do*".

Where direct imitation was absent, several participants described indirect cross-sector learning. P2 cited a competitor as a reference: "*In terms of direct comparisons, [Anonymised] operates on a somewhat similar model with a centralised hub. But the idea of truly putting customers at the centre of operations, I have not seen that much elsewhere*". P5 described a cross-industry benchmarking process: "w*e also looked at what other brands were doing, not so much in the beauty sector specifically but brands simply positioned in the sustainable development space more broadly. We saw how they operated, and then we just looked, compared, and figured out how it could work with electric bikes and cargo bikes*".

P1 acknowledged a broader dynamic: "*for all companies there is inevitably a kind of cascade effect that has impacted us*". Furthermore, two participants demonstrated acute competitive awareness even while denying peer imitation. P4 refused to share advice with competitors:

"*Not at all especially not to competitors. You don't give advice to competitors*"; a stance that reveals strategic consciousness of competitive position. P8 openly acknowledged competitor observation: "*for daytime delivery, however, there are always new players entering the market, and it is possible for us to integrate them or observe what they do and see if it can be proposed to our current carriers*".

The temporal dimension also matters. P2 noted: "*We tend to do that now, but honestly at the beginning, no*" suggesting that peer observation may emerge after adoption rather than driving it. P7 confirmed: "*especially when we started thinking about it during Covid, it was not something that was very widespread at the time*". P9 took the strongest anti-mimetic position: "*Honestly, yes a definite yes. [...] 10 years ago I would have done exactly the same thing*".

4.3.4 Economic considerations

Economic factors played a dual role across the sample: as both a barrier and a motivation, depending on the company's size and strategic horizon.

At Case 1, P1 identified a direct economic incentive: "*Last Mile delivery accounts for 70 to 80% of the total cost of delivering a product. [...] micromobility and the [Electric van] allow us to deliver more parcels more often at a lower cost*". P1 also framed electrification as a hedge against geopolitical risk: "*the fact that oil prices risk exploding due to tensions in the Middle East is a factor for all logistics chains worldwide. And we are glad to have taken the lead and already implemented several electric and micromobility solutions to reduce these dependencies on oil*".

P2 acknowledged the cost tension: "*Sustainable delivery is more expensive. That was another challenge we faced, but one we managed to overcome*". However, P2 also noted the economic benefit of electric vehicles given rising fuel prices: "*We are very happy about that, especially given the rising price of petrol*". P2 described a concrete operational gain from route optimisation: "*We ran two changes: now when we make a delivery, we collect their previous jars at the same time. [...] I think today things have really changed, I believe we gained around 10% in performance because of these two changes*".

P4 highlighted long-term savings: "*What also motivated us a lot at the start was fuel cost. Doing 300 km in a single day, at some point that adds up quite a bit. So that is partly why we did not really hesitate and invested directly in an electric van*".

P6 explicitly cited the economic logic of relay-point delivery: "*These solutions are interesting from several perspectives, both financially for the company and for the customer, in the sense that the final price the customer pays is lower than home delivery*". P6 further connected this to the macroeconomic context: "*it allows for lower prices, which today is a strong argument given inflation and the state of the economy where we need to stimulate consumption through lower prices*".

P10 was the most transparent about cost being the primary driver: "*To be honest, one of the first motivations is cost. By consolidating and optimising routes, we limit the costs of temperature-controlled delivery, which is very expensive due to the energy requirements*". P10 also described a dual ecological-economic logic in avoiding empty trips: "*We also try to avoid empty trips, if I send a carrier to a client in Boulogne, I try to organise a collection in that same area for the return journey. It is both more ecological and more cost-effective*". The geopolitical context added urgency: "*The recent geopolitical context with the conflict in Iran has pushed up transport costs significantly, which is forcing us to rethink*".

P8 highlighted the prohibitive investment dimension: "*The cost of an electric truck is extremely high. So we have to calculate how we are going to amortise this cost over 2 to 3 years*". P8 explained the corporate framework behind this: "*often in companies, when a project is launched, large companies require a return on investment within 3 years. So the analysis was done in that sense*".

P3 described a survival versus sustainability trade-off faced by some client brands: "*Do you want to keep your business alive, or do you want to invest in CSR? The question can be asked today. I don't feel that CSR is something that is prioritised over survival*". P3 also acknowledged the cost pressure on transport: "*companies are under pressure on their transport costs. They want to pull costs down. If a company doesn't have the cash flow or the budget margin to invest in this, they will prefer to cut costs*". However, P3 noted that green solutions could also serve as a hedge: "*if on the last mile we can anticipate by using electric vehicles or biofuels, it helps to absorb a bit of those cost surges*".

4.3.5 Government regulation and policy

Government regulation emerged as the weakest explicit motivator in the sample. The majority of participants, particularly those in smaller companies, did not identify regulation as a primary driver.

P2 stated: "*That is more of a minor factor [...] it is not really a driver for us*". P4 was even more emphatic: "*Even without any rules, we would have adopted the electric van anyway*". P9 was philosophically opposed to the regulatory framing: "*I prefer to be small, autonomous, responsible and ecological on my own, and make the government bend eventually because it is logical*". In addition, P6 reported: "*Not at our level, at least at my level, we do not feel them much*". And P11 confirmed: "*We are not large enough to be directly concerned by government regulations*".

For larger companies, the picture was more nuanced. P8 described regulation as the carriers' concern rather than the company's: "*Carriers, on their side, know that there are more and more demands from governments and regulations on these topics. So they either have to take the lead or comply when regulations are implemented. [...] today that is mainly the role of carriers, not ours, since we do not own transport operations*".

P1 acknowledged fiscal incentives and local partnerships: "*I know we are also in partnership with the city of Munich. We are obviously in constant contact with local officials to find the broadest possible solutions, with a favourable tax framework and integration into the local landscape*". Unlike most other participants, P1 considered governments regulations as a strong driver.

The anticipatory dimension was more present than direct compliance. P10 explained: "*We use a B2B platform called [Anonymised] where we have to fill in all company information including a full RSE section. [...] We want to be as clean as possible in those documents so that sustainability requirements do not block future commercial relationships*". P10 made the anticipatory logic explicit: "*More and more corporate clients will require us to go through that kind of platform. [...] So it is anticipatory, we are trying to get ahead of regulations that may tighten in the future, even if they do not directly apply to us today*".

P3 also adopted a similar preparedness logic: "*Tomorrow, the idea is not to be caught off guard by government restrictions, we have the solution. If a client comes to us and says 'OK, I have*

no choice in this country, I have to deliver this way', no problem, we have the right solutions, we'll activate them". He further described a deliberate infrastructure for this preparedness: "*we have the benefit of working with a huge number of partners and our list of available services is very long. [...] what [Anonymised]does to prepare for tomorrow, meaning stricter restrictions, is to have backup solutions ready to activate*".

P8 described a proactive carrier-challenging approach aligned with regulatory horizons: "*On the France network, I think it is through that approach that we challenge our carriers. That's why we ask them what they are implementing, to be ahead of major deadlines like 2030 or 2050*".

P11 connected anticipatory compliance to growth ambitions: "*We are ambitious and obviously want to grow as much as possible. So given the possibility that at some point [...] we could be concerned by these laws, yes, we are glad that we are already aligned*". He further predicted: "*In any case, given what has been happening over the past few years, it would not be surprising if in the fairly near future sustainable deliveries became mandatory for everyone. It is therefore better to get on the train now*".

4.4 The role of the urban context

Participants described the urban environment as both a facilitator and a constraint for sustainable last-mile implementation.

4.4.1 The urban context as enabler

Several participants highlighted that urban infrastructure and density made sustainable solutions viable. P5 noted: "*Since the offices are in central Paris, there is a lot of passage, a lot of routes, a lot of cycle lanes. For cargo bikes, it is not complicated to come and collect the goods from the boutique and deliver them to customers*".

P9 described how local infrastructure directly enabled the cargo bike model: "*The last mayoral term was really focused on developing cycle paths. So honestly, everything is set up to make the cargo bike solution very easy to implement. The cycle lanes are wider. They serve almost the whole town*".

P11 made a general observation: "*In general, the denser the urban context, the more sustainable delivery solutions are available. In Paris again, the solutions are very numerous and I think cargo bikes are going to take a really significant share of the Parisian market*".

P8 provided a specific example from the Paris Olympic Games: "*There were many restricted zones, red zones, and what the city did was create dedicated parking areas for delivery drivers. [...] They would park and complete the last part of the delivery on foot [...] It was an ecological approach and it worked very well*". He also noted the importance of city-specific infrastructure: "*some cities are more suitable for implementing bike deliveries because they have bike lanes, because the routes are easier to optimise, or because they are smaller cities where local agencies are closer to the city centre, making round trips easier*". P8 further highlighted a distinctive feature of the company's network: "*we also have a major advantage, which is that the majority of our network operates at night. So we do a lot of deliveries to opticians via mailboxes, which allows us to deposit goods. That means we can drive faster because there are fewer people on the road*". This night delivery model is unique in the sample and represents an alternative approach to urban congestion.

Case 2 described municipal support as an enabler: "*The city is even supporting us financially in that regard. [...] I think we are genuinely confident about the future. As I said, the city is behind us and with us*".

P3 provided quantitative estimates of what urban infrastructure makes possible: "*for these clients who are very sensitive to CSR and last-mile green solutions, I think between 50 and 60% of their deliveries can actually be green*". He further estimated: "*I think it can go a bit higher and we can be closer to around 60-70% when we're talking about metropolises and major cities*".

4.4.2 The urban context as constraint

Conversely, urban density created specific operational constraint. P1 described the tension between vehicle size and city streets: "*The [Electric van] are larger than conventional vans. [...] I'm thinking for example of Amsterdam, where we carry out a very large share of our deliveries via micromobility because the vans simply cannot pass through most of the streets*". P1 described a design challenge caused by European urban infrastructure: "*the size of the US vans, given local infrastructure, was much larger than most European cities could*

accommodate. So we had to revise their development to reduce their volume so they could fit through most of our routes".

P10 highlighted the complexity of Parisian logistics: "*Paris is genuinely very complex logistically. Every time we need a delivery, there is a risk of getting stuck in traffic. [...] clients often have strict delivery bay restrictions, which means I have to provide the driver's identity and vehicle registration number in advance*". He also identified a product-specific urban constraint: "*In summer, it is simply impossible to ship chocolate via a standard carrier like Colissimo, because if a parcel spends two or three days in transit, possibly sitting in places where temperatures can reach 26 to 35 degrees, it is absolutely impossible to leave it at ambient temperature*". This seasonal heat constraint illustrates how urban density, combined with product characteristics, creates delivery challenges absent in non-perishable sectors.

P8 noted the trade-off between bike capacity and van access: "*You can carry many more parcels in a van than on a bike, so we need to have the ability to go back and forth to the sorting centre or the local agency in order to maintain the same delivery standards as with vans*".

Case 3 pointed to the urban-rural divide: "*If it's a small town, somewhere in the middle of nowhere, it's a bit more complicated*". P3 also identified a reverse logistics challenge: "*our warehouses are not in city centres, it is harder to deliver goods by low-emission vehicle back to the warehouse. So finding sustainable return solutions is more complex*". This observation highlights that the urban context enables sustainable last-mile delivery but does not necessarily support the return of the supply chain.

The data suggests that urban constraints affect different sustainable solutions differently. Electric vehicles face challenges related to vehicle size, parking and charging infrastructure, issues highlighted by P1 regarding the electric vans and by P8 regarding the limited availability of charging stations. Cargo bikes, by contrast, appear to benefit from urban density: cycle lanes, pedestrian zones and narrow streets that constrain vans become advantages for bike-based delivery. P7 noted that "*in central Paris, it is faster*" by cargo bike. P11 observed: "*When you walk around the city, you can see that bikes move much more fluidly than vans. They pollute less and are far more flexible than other vehicles*". P9 confirmed that the cargo bike's operational range is the binding constraint: "*I admit that if we go beyond about 20 km, an electric vehicle solution would become viable, because you also have to be practical. You cannot spend four hours delivering by bike*". This pattern suggests that the urban context

functions as a constraint primarily for EV-based solutions, while simultaneously enabling cargo bike delivery.

5. Analysis

This chapter presents the second level of analysis, in which the empirical data described in Chapter 4 is examined through a cross-case comparison using the three mechanisms of Institutional Theory: coercive, mimetic and normative isomorphism as sensitising concepts. The analysis identifies the patterns, convergences and tensions that emerged across the eleven cases and addresses the research question: what institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies in urban contexts.

Chapter 4 established that participants described a diverse range of motivations, competitive dynamics and urban conditions shaping their adoption decisions. This chapter now examines these empirical patterns through the analytical lens of Institutional Theory. Before presenting the detailed cross-case analysis, Figure 4 provides an overview of how the three institutional pressures manifested across the eleven cases, grouped by firm size. This comparison highlights the structural differences in how micro-enterprises, mid-size companies and large companies experience and respond to institutional pressures, a pattern that runs through the entire analysis that follows.

Final Theme	Micro-enterprises (Cases 4, 6, 9, 11)	Mid-size (Cases 2, 5, 7, 10)	Large (Cases 1, 3, 8)
Normative: internal values	Founder imprint direct, non-negotiable	Brand positioning, mediated through strategy	Corporate CSR departments, training
Normative: customer expectations	High, direct B2C contact	High but selective (product > delivery)	Low (speed priority)
Coercive: regulation	Irrelevant	Anticipatory	Delegated to carriers, 2030/2050 horizons
Coercive: urban context	Enabler (cycle lanes, density)	Mixed	Constraint (vehicle size, parking)
Mimetic	Absent (no visible peers)	Indirect (cross-sector)	Acknowledged (competitor scanning)
Economic dimension	Low gate (cargo bike cheap)	Medium (absorbed internally)	High gate (3-year ROI)

Figure 4: Cross-case comparison of institutional pressures by firm size

5.1 Normative Institutional pressures: the dominant mechanism

The most striking finding across the eleven cases is the dominance of normative pressures as the primary institutional force shaping sustainable last-mile adoption. In the present sample, normative pressures operated through two distinct channels.

The first channel is internal normative pressure, driven by the founding values and identity commitments of the organisation. This was the dominant pattern among SMEs and micro-enterprises. P9 stated that the cargo bike "*was obvious*" and that the question of alternative delivery methods "*did not even arise*". P4 confirmed that sustainable delivery was adopted "*even without any rules*" because it was "*part of our company values*". P5 described the alignment between delivery and brand identity as a non-negotiable imperative. P9 provided the most vivid formulation of this imperative: "*It was not even a question. I could not imagine making a super virtuous broth and delivering it with an anti-virtuous solution. It simply did not fit the company's values at all*". The term "anti-virtuous" captures the logical impossibility, as perceived by the founder, of dissociating the product from its delivery. These accounts suggest that for identity-driven companies, sustainable delivery is not a strategic choice among alternatives but a constitutive element of the organisational self-concept. The data suggests that in the context of sustainable logistics, founder-level values and brand identity function as powerful normative forces, particularly in smaller organisations where the distance between founder values and operational decisions is minimal.

This finding reveals a clear firm-size effect. In micro-enterprises (Cases 4, 6, 9, 11), the founder's personal values directly determined the delivery model. In mid-size companies (Cases 2, 5, 7, 10), values remained central but were mediated through organisational processes brand positioning, supplier selection and carrier negotiations. In large companies (Cases 1, 3, 8), normative pressure operated through corporate strategy, dedicated CSR departments and training programmes rather than through personal founder conviction. The imprint of the founder weakens as organisational size increases, but the normative mechanism persists, it simply operates at a different organisational level.

Importantly, this internal normative channel was not limited to small companies. P8 described the transition as "*chosen. We are not forced, these are decisions made by the company, even if they create challenges*". This suggests that even in large corporate environments where

multiple institutional pressures coexist, the voluntary dimension of adoption is perceived as primary.

The second channel is external normative pressure, driven by evolving customer expectations and broader societal change. P3 described "*societal change*" as the most important driver, explaining that sustainability "*is becoming the norm*" and that "*future employees and company directors will have it built into their habits*". P2 related a customer base that would actively punish regression: returning to thermal vehicles "*would definitely make us lose customers*". P2 provided the only quantitative evidence in the sample: a survey of approximately 1,000 customers found that around 75% were willing to pay more for sustainably sourced products. This empirical measurement suggests that customer expectations are not merely perceived but demonstrated.

However, the findings also reveal a gap between stated customer preferences and actual behaviour. P11 observed: "*when delivery is not included in the price, customers tend to choose the least ecological delivery option because it is the cheapest*". P5 noted: "*customers, even if they consider the product to be good for health, do not necessarily see the other side of things*". These observations suggest that external normative pressure from customers operates more as a reputational risk (what happens if we stop) than as an active purchasing criterion (customers choose us because of delivery sustainability).

The interaction between internal and external normative pressures creates what can be described as a normative coherence imperative: once a company establishes a sustainability-oriented identity, every operational decision, including delivery, must be aligned with that identity to maintain legitimacy with both internal stakeholders and external customers. P5 captured this most clearly: "*We cannot sell products and deliver them in diesel trucks*". P11 designed an entire delivery pricing model around this logic. P11 articulated the risk of incoherence: "*If our discourse is disconnected from our actions, then yes, I think it would affect sales, because it is hard to inspire trust in customers that way*". This formulation connects the normative coherence imperative directly to commercial outcomes.

5.2 Coercive Institutional pressures: present but indirect

Coercive pressures were less explicitly cited by participants than normative pressures. Most participants, particularly those in smaller companies, did not identify government regulation as a primary driver of their adoption decisions.

However, closer analysis reveals that coercive pressures are more present than initial accounts suggest, but they operate indirectly rather than through direct compliance. Four indirect channels were identified.

First, anticipatory compliance. Several companies were not responding to current regulations but positioning themselves for expected future tightening. P10 described filling sustainability documentation on B2B platforms to avoid being "*blocked*" by future commercial requirements. P10 made this logic explicit in their own words: "*So it is anticipatory, we are trying to get ahead of regulations that may tighten in the future, even if they do not directly apply to us today*". P3 framed preparedness as a strategic asset: the goal is "*not to be caught off guard*". P8 described a proactive carrier-challenging approach aligned with regulatory horizons: "*On the France network, I think it is through that approach that we challenge our carriers. That's why we ask them what they are implementing, to be ahead of major deadlines like 2030 or 2050*". This connects anticipatory compliance to specific regulatory timelines, showing that even when companies do not experience direct coercive pressure, distant regulatory targets shape present behaviour.

Second, delegated coercive pressure. For companies that outsource delivery, regulatory pressure is absorbed by carriers rather than by the e-commerce company itself. P8 stated explicitly: "*That is mainly the role of carriers, not ours, since we do not own transport operations*". This creates a structural buffer: the company can adopt sustainable delivery without experiencing coercive pressure directly, because the carrier has already adapted to regulatory requirements. Case 1 presents the inverse of this mechanism. Rather than delegating coercive pressure to carriers, this company imposes its own fleet on delivery partners: "*we have a defined set of vehicle types that we offer to our DSPs, Delivery Service Partners [...] So at certain Delivery Stations we only use [Electric van], and in that case we provide them to our delivery partners*". In this configuration, the e-commerce company is the source of coercive pressure, not the recipient. This reversal suggests that the direction of coercive pressure depends on the power asymmetry between the e-commerce company and its logistics partners.

Third, urban context as soft coercion. While most participants did not describe low-emission zones or access restrictions as primary drivers, several acknowledged that the urban environment created practical incentives to adopt sustainable solutions. P4 framed urban restrictions as enabling market access: being electric means "*not having to say no to future clients*". P1 linked Amsterdam's narrow streets directly to the dominance of micromobility. These pressures are not formal regulations in the traditional sense, but they function coercively by constraining the operational space available for conventional delivery.

Fourth, normative-to-coercive conversion emerged. P10 described how B2B sustainability platforms effectively transform voluntary sustainability norms into quasi-regulatory requirements: "*More and more corporate clients will require us to go through that kind of platform*". Similarly, P3 described responding to client requests for green delivery as if they were mandatory constraints. In these cases, what begins as a normative expectation from a commercial partner acquires coercive force through its binding effect on the business relationship.

The relative weakness of direct coercive pressure in this sample is consistent with the composition of the sample itself: the majority of companies are SMEs below the regulatory thresholds that trigger mandatory sustainability reporting or fleet conversion requirements. For the three larger companies (Cases 1, 3 and 8), coercive pressures were more visible but still mediated by corporate structure, carrier relationships and local variation.

5.3 Mimetic Institutional pressures: the weakest mechanism

Mimetic pressures were the least observable mechanism across the eleven cases. Most participants reported limited direct observation of or influence from competitors' sustainable delivery practices. P4 stated: "*We don't really know any direct competitors*". P7 said: "*I am not sure. Personally, I do not know many similar cases*". P5 noted: "*That was a fairly minor aspect*". Several explanations emerge from the data.

First, many companies in the sample are first-mover operators in their specific segments. Case 4 operates in vintage crockery, Case 6 in second-hand children's books, Case 9 in artisanal broth. In such fragmented markets, there are few visible peers to imitate.

Second, where mimetic processes were present, they operated indirectly. P10 drew inspiration not from direct competitors but from "*wine distributors who share our logistics provider*", P5 described looking at "*brands simply positioned in the sustainable development space more broadly*" rather than beauty competitors specifically. P3 acknowledged that management attended "*external trade shows and events where companies come together, so there must have been some influence*". P2 cited a competitor as a cross-sector reference: "*In terms of direct comparisons, [Anonymised] operates on a somewhat similar model with a centralised hub*". P8 was the only large company to openly acknowledge competitor observation: "*that does not prevent us from taking inspiration from our competitors*" and described scanning new market entrants: "*there are always new players entering the market, and it is possible for us to integrate them or observe what they do*". These findings suggest that in fragmented industries, mimetic pressures may operate across sectoral boundaries rather than within them.

Third, the temporal dimension reveals that peer observation often follows adoption rather than driving it. P2 stated: "*We tend to do that now, but honestly at the beginning, no*". P7 confirmed: "*especially when we started thinking about it during Covid, it was not something that was very widespread at the time*". P9 took the strongest anti-mimetic position: "*Honestly, yes a definite yes. [...] 10 years ago I would have done exactly the same thing*". These accounts suggest that mimetic awareness is a consequence of field maturation rather than a driver of initial adoption.

The weakness of mimetic pressures in the data does not mean they are absent from the institutional field. Rather, their effects may be deferred or masked: what participants describe as "*societal change*" or "*the norm*" (P3) may in fact reflect the cumulative outcome of mimetic processes operating at the field level, even if individual firms do not consciously attribute their decisions to competitor observation. P1 came closest to describing this dynamic: "*for all companies there is inevitably a kind of cascade effect that has impacted us*". This "cascade effect" may be the clearest articulation of field-level mimetic dynamics in the sample a diffuse, collectively generated pressure that no single actor claims to have initiated.

However, Case 6 represents a partial exception. P6 identified competition as the most important driver and explicitly stated that the ecological business model "*was indeed somewhat motivated by a desire for differentiation relative to those competitors*". P6 was not copying a specific competitor's delivery model but rather positioning against the field; it does suggest awareness of and responsiveness to the broader competitive landscape. This case may therefore represent

an intermediate category between competitive strategy and mimetic alignment, where the firm responds not to a specific model but to a perceived direction of the organisational field.

5.4 Interactions between mechanisms

A key finding of this study is that the three institutional mechanisms do not operate in isolation but interact in specific configurations depending on company size and organisational model.

For micro-enterprises with sustainability in their DNA (Cases 4, 6, 9, 11) the dominant configuration is: strong normative (internal values) → reinforced by normative (customer expectations) → minimal coercive → absent mimetic (except for case 6). Sustainable delivery is adopted because it is constitutive of the firm's identity, not because of external pressure. In these cases, the founder and management's imprint on the adoption decision is direct and traceable the delivery model is an extension of their personal values and lifestyle choices.

For mid-size companies with sustainability as a differentiator (Cases 2, 5, 7, 10), the configuration shifts to: normative (values + customers) + competitive positioning → some anticipatory coercive → limited mimetic (cross-sector). These firms adopted sustainability partly for identity reasons and partly for strategic differentiation, and they show greater sensitivity to future regulatory developments.

For large companies (Cases 1, 3, 8) the configuration is: mixed normative (corporate commitments, societal norms) + delegated coercive (through carriers and municipalities) + implicit mimetic (field-level normalisation) + strong economic calculation. These companies operate in more complex institutional environments where multiple pressures interact simultaneously and where the decision-making process involves more actors and longer time horizons.

A cross-cutting pattern across all three configurations is the role of the economic dimension as a mediator rather than a standalone driver. In the present sample it functions as a gate: normative commitment must pass through economic feasibility before it translates into adoption. P3 captured this tension directly: "*Do you want to keep your business alive, or do you want to invest in CSR*?" For micro-enterprises, the economic gate is lower (a cargo bike costs less than an electric truck); for large companies, the 3-year ROI requirement described

by P8 creates a higher threshold. This suggests that the relative strength of institutional pressures is mediated by the economic capacity of the firm.

5.5 An emergent theme: the normative coherence imperative

A theme that emerged from the data but is not fully captured by the three isomorphism mechanisms is what can be termed the normative coherence imperative: the organisational pressure to ensure that every element of the value chain, including delivery, is aligned with the company's stated identity. This is distinct from external normative pressure because it originates from within the organisation and is self-imposed. It is also distinct from rational strategic calculation because participants described it not as a deliberate choice but as an imperative: "*I simply could not imagine delivering any other way*" (P9); "*Obviously we cannot sell products and deliver them in diesel trucks*" (P5).

This finding is consistent with the abductive logic of this study, which remains open to themes that "may enrich, nuance, or challenge the framework" (Section 3.5). The normative coherence imperative suggests that for identity-driven firms, the boundary between normative pressure and organisational identity is blurred: the firm does not conform to an external norm but rather generates its own normative standard from its values and then experiences that standard as binding.

However, the data also suggests that normative coherence is a spectrum rather than a binary state. P5 acknowledged: "*there is still a real margin to make the whole brand green, truly sustainable. We are really more focused on the products than on the delivery*". This suggests that even within companies that experience the normative coherence imperative, full alignment between product sustainability and delivery sustainability is an ongoing process. The imperative creates directional pressure, but its realisation depends on resources and operational constraints.

6. Discussion and conclusion

This chapter discusses the contributions of the study to both Institutional Theory and the broader literature on sustainable last-mile delivery. It then presents the practical and societal implications, acknowledges the limitations and proposes directions for future research.

6.1 Contribution to Institutional Theory

This study set out to explore what institutional pressures shape the adoption of sustainable last-mile delivery solutions by e-commerce companies in urban contexts. The analysis of eleven cases through the lens of Institutional Theory reveals that normative pressures, particularly internal values and customer expectations, are the dominant institutional force shaping adoption across the sample. Coercive pressures operate indirectly through anticipatory compliance, carrier-level delegation and urban infrastructure constraints. Mimetic pressures are the weakest mechanism observed, partly due to the fragmented nature of the industries studied and the niche positioning of most firms.

The study identifies an emergent theme, the normative coherence imperative, which describes the self-imposed pressure on identity-driven firms to align every element of their value chain with their stated sustainability commitments. This mechanism extends DiMaggio and Powell's (1983) original framework by highlighting the role of organisational identity as a source of normative pressure. Unlike the external normative pressures described by DiMaggio and Powell (1983) which arise from professionalisation and field-level cognitive frameworks the normative coherence imperative originates from within the organisation. It suggests that for values-driven firms, the firm does not conform to an external norm but generates its own normative standard and then experiences that standard as binding.

The study also identifies two novel mechanisms not previously documented in the literature on sustainable logistics adoption. First, reverse coercion, where a dominant e-commerce platform (Case 1) imposes sustainable fleet requirements on its delivery service partners, inverting the typical direction of coercive pressure. Second, normative-to-coercive conversion, where

voluntary sustainability norms become quasi-mandatory through B2B platform requirements and commercial dependency.

Finally, the analysis reveals that the three isomorphism mechanisms interact in distinct configurations depending on firm size: micro-enterprises are driven almost exclusively by internal normative pressures with direct founder imprint; mid-size companies combine normative with competitive and anticipatory coercive pressures; and large companies experience a more complex mix where economic calculation, delegated coercion and implicit mimetic dynamics coexist. This firm-size effect suggests that the relative weight of institutional pressures is not fixed but varies systematically with organisational scale.

6.2 Contribution to the literature on sustainable last-mile delivery

The findings both confirm and nuance patterns documented in the existing literature. The identification of regulatory pressure, competitive dynamics and customer expectations as factors shaping sustainable last-mile adoption is consistent with the multi-level driver framework identified in the literature review (Section 2.1.2.1). The budget constraints reported by SMEs confirm the financial barriers documented by Dadashzada et al. (2025) and Thoma and Gruber (2020). The speed-sustainability tension described by P8 and P5 is consistent with Rose et al. (2025), who argue that consumer expectations for speed constrain the operational space for sustainable adoption. The enabling role of urban infrastructure documented in Section 4.4.1 is consistent with Chatziioannou et al. (2024) and Castrellon and Sanchez-Diaz (2024).

However, the findings nuance the literature in several important ways. First, while regulatory pressure is consistently identified as one of the most powerful drivers in the literature (Paddeu, 2025; Valentini et al., 2026), this study finds it to be among the least visible drivers across the sample. This discrepancy is explained by the composition of the sample (predominantly SMEs below regulatory thresholds) and by the delegation mechanism: in outsourced delivery models, regulatory pressure is absorbed by carriers rather than experienced by the e-commerce company.

Second, while the literature treats normative pressures as external professional norms and societal expectations (Tetteh et al., 2024; Marculetiu et al., 2023), this study identifies a

powerful internal dimension: the normative coherence imperative, where firms generate their own binding normative standards from their founding values.

Third, the temporal evidence suggests that mimetic awareness often follows adoption rather than driving it. P2 stated that peer observation began "*now, but honestly at the beginning, no*". This reverses the causal direction implied in DiMaggio and Powell's (1983) framework and suggests that in emerging fields, mimetic processes may be a consequence of field maturation rather than a driver of initial adoption.

Fourth, the finding that customer expectations function more as a reputational risk than as an active purchasing criterion, evidenced by P11's observation that customers choose the cheapest delivery when given the choice, nuances the literature on consumer-driven sustainability (Tetteh et al., 2024). The structural mechanisms adopted by Cases 6 and 11 to pre-empt this behaviour represent organisational solutions to a gap between stated preferences and actual consumer behaviour that the literature has documented (Rose et al., 2025) but not yet linked to delivery model design.

The study also extends the literature by documenting the free-delivery model (Case 11) and the included-in-price relay-point strategy (Case 6) as deliberate organisational design choices that remove the sustainability decision from the consumer and embed it in the company's pricing structure. P6's distinction between "offered" and "free" delivery reveals a behavioural framing mechanism not previously documented through qualitative interviews with the firms that implement them.

6.3 Practical & Societal implications

The findings carry implications for multiple stakeholders. For managers and entrepreneurs, the study suggests that sustainable delivery should be considered not as a supplementary cost but as a constitutive element of brand identity that builds customer trust and long-term competitive advantage. Cases 2, 4, 9 and 11 demonstrate that even small companies with limited budgets can implement sustainable delivery when it is embedded in the business model from the start.

For policymakers, the finding that government regulation is the weakest explicit driver does not mean that policy is irrelevant. The anticipatory compliance pattern suggests that the mere

signalling of future regulatory tightening can influence firm behaviour, even before regulations are enforced. The specific reference to 2030 and 2050 deadlines by P8 confirms that distant regulatory targets already shape present-day carrier management decisions. Furthermore, the enabling role of urban infrastructure (cycle lanes, charging points, dedicated delivery zones) was highlighted by multiple participants as a practical facilitator of sustainable adoption. P2 described direct municipal financial support, P9 highlighted comprehensive cycle lane networks, and P3 estimated that urban infrastructure enables 60-70% green delivery rates in major cities. This suggests that enabling urban infrastructure may be a complementary lever to regulation in supporting sustainable adoption.

For logistics service providers and carriers, the study reveals that e-commerce companies increasingly use sustainability credentials as a selection criterion in carrier tenders. The pressure described by P3 to provide concrete evidence of green last-mile practices suggests a growing demand for transparency and verifiability in sustainable logistics services. However, P7 noted that carriers "*do not proactively inform us of what our impact has been*", suggesting that the supply of sustainability data from carriers lags behind the demand from e-commerce companies. Closing this information gap represents both a commercial opportunity and a policy lever.

6.4 Limitations

This study has several limitations that should be considered when interpreting the findings.

First, the sample consists predominantly of French companies operating in urban contexts shaped by specific regulatory, infrastructural and cultural conditions. France has a particularly developed relay-point network, active municipal policies promoting cycling infrastructure and pedestrianisation and an urban regulatory environment characterised by expanding low-emission zones. These factors may have amplified the role of the urban context as a facilitator of sustainable adoption in ways that would not replicate in contexts with less developed infrastructure. The single German case (Case 1, Munich) provides limited comparative perspective. Transferability to non-European contexts should therefore be treated with caution.

Second, the reliance on a single respondent per organisation represents a limitation for a study using Institutional Theory as its framework. Institutional pressures operate at the organisational

level, yet each case reflects the perceptions of one individual, whose position and personal values may shape how they frame the company's motivations. This is particularly relevant for the normative pressures identified in this study: a founder who built sustainability into the company's DNA may describe normative coherence as self-evident, while an operational manager hired later may perceive the same decision as more strategically motivated.

Third, the sample spans a wide range of organisational scales, from micro-enterprises with fewer than five employees to multinationals with global operations. While this diversity enables cross-case comparison, it also means that the institutional mechanisms observed may not be fully comparable. The findings for large companies should be read as illustrative rather than as direct equivalents to the SME findings.

Fourth, the written response format used for P11 produced systematically shorter and less dialogic data than the live interviews. Additionally, written responses may reduce spontaneity and allow participants more time to craft answers aligned with their public brand positioning.

Fifth, the study relies exclusively on self-reported data without triangulation from secondary sources such as sustainability reports, CO2 emissions data or carrier contracts. Several participants acknowledged the absence of quantitative sustainability metrics within their own operations, making it impossible to verify whether the practices described match actual environmental outcomes.

Finally, the translation of eleven interviews from French to English introduces a risk of semantic loss. While the research team reviewed all translations carefully and used AI assistance to maintain consistency, idiomatic expressions, hedging language and tonal nuances, all of which are analytically relevant in qualitative research may have been partially altered in the process.

6.5 Suggestions for future research

Several directions for future research emerge from these findings. First, a larger-scale study could test whether the dominance of normative pressures observed here holds across a broader and more diverse sample, including non-European contexts where regulatory frameworks and cultural attitudes toward sustainability may differ significantly. Second, longitudinal research

could examine how the relative weight of institutional pressures evolves over time as regulations tighten and sustainable delivery becomes more normalised. The temporal evidence in this study, with several participants noting that sustainability awareness has increased significantly over the past decade suggests that the relative strength of the three isomorphic mechanisms may shift over time, with mimetic processes potentially gaining strength as the field matures and visible models become more numerous. Third, future studies could investigate the consumer side and search how do end consumers perceive and value sustainable delivery? and to what extent do their stated preferences translate into actual purchasing behaviour? The gap between stated preferences and actual choices identified in this study (P11, P5) points toward a behavioural research agenda that could complement the institutional perspective adopted here.

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Appendix 1: Interview questions

Bloc 1 - Introduction & context

Q1. Could you briefly introduce yourself and describe your role within the company?

FR : Pourriez-vous vous présenter brièvement et décrire votre rôle au sein de l'entreprise ?

Q2. Could you give us a brief overview of your company's last-mile delivery operations (vehicules), particularly in urban areas?

FR : Pourriez-vous nous donner un aperçu des opérations de livraison du dernier kilomètre de votre entreprise, notamment dans les zones urbaines ?

Bloc 2 - The adoption decision

Q3. What sustainable last-mile delivery solution(s) has your company adopted? when did this transition took place?

FR : Quelles solutions de livraison durable votre entreprise a-t-elle adoptées ? quand cette transition a-t-elle eu lieu ?

Q4. How did you proceed to find a sustainable solution that matches your needs?

FR : Comment avez-vous procédé pour trouver une solution durable qui répondait à vos besoins ?

Q5. What were the main motivations behind this decision, was it driven by regulation, competitive pressure, customer expectations, internal values, or something else?

FR : Quelles étaient les principales motivations derrière cette décision, était-ce lié à des réglementations, à la pression concurrentielle, aux attentes clients, aux valeurs internes, ou autre chose ?

Q6. Did the urban context specifically, such as access restrictions, low-emission zones, or city regulations, play a role in this decision?

FR : Le contexte urbain spécifiquement, comme les restrictions d'accès, les zones à faibles émissions, ou les réglementations municipales, a-t-il joué un rôle dans cette décision ?

Q7. Were you influenced by what other companies in your sector were doing in terms of sustainable delivery?

FR : Avez-vous été influencés par ce que d'autres entreprises de votre secteur faisaient en matière de livraison durable ?

Q8. How did the transition unfold operationally? Were there significant challenges during the implementation phase?

FR : Comment s'est déroulée la transition sur le plan opérationnel ? Y a-t-il eu des difficultés importantes durant la phase de mise en place ?

Q9. How did the specific characteristics of the urban environment, such as traffic density, infrastructure, or delivery access, affect the implementation of your sustainable solution?

FR : Comment les caractéristiques spécifiques du contexte urbain, comme la densité du trafic, l'infrastructure ou les accès de livraison, ont-elles affecté la mise en œuvre de votre solution durable ?

Bloc 3 - Hindsight & broader perspective

Q10. Looking back, how would you describe your overall experience of this transition, what went well? and what would you do differently?

FR : Avec le recul, comment décririez-vous votre expérience globale de cette transition, qu'est-ce qui s'est bien passé ? que feriez-vous différemment ?

Q11. Do you feel that the urban context made this transition more necessary, more difficult, or both?

FR : Estimez-vous que le contexte urbain a rendu cette transition plus nécessaire, plus difficile, ou les deux ?

Q12. What advice would you give to an e-commerce company that is currently hesitating to make a similar transition?

FR : Quel conseil donneriez-vous à une entreprise e-commerce qui hésite actuellement à faire une transition similaire ?

Q13. Is there anything else you would like to add that we have not covered and that you feel is relevant to this topic?

FR : Y a-t-il autre chose que vous souhaiteriez ajouter que nous n'avons pas abordé et que vous jugez pertinent sur ce sujet ?

Appendix 2: Thematic Coding Table

	A	B	C	D
1	**Case Number**	**1st Order Codes**	**2nd Order Codes**	**Final Themes**
2	Case 2	*"The core idea is to offer food products in glass jars, to give customers that eco-responsible consciousness"*		
3	Case 2	*"We could not neglect the operational and delivery side. That is why we transitioned to electric vehicles"*		
4	Case 2	*"Going back? Yes, that would definitely make us lose customers, people who no longer trust us and feel our values have changed"*		
5	Case 4	*"Even without any rules, we would have adopted the electric van anyway, because it is part of our company values"*		
6	Case 4	*"The main reason was above all that it is much more aligned with the fact that we work with second-hand items. The whole idea of not wasting, of circular economy"*		
7	Case 5	*"It immediately comes down to aligning with the brand's values. [...] we position ourselves as green, as a brand that truly advocates responsible beauty. So obviously we cannot sell products and deliver them in diesel trucks"*		
8	Case 6	*"We made a fairly deliberate choice across the entire value creation chain, from sourcing to final delivery to customers, to have a strong ecological DNA that is really part of the business model"*		
9	Case 7	*"The idea of implementing something in line with that vision and mindset came directly from the founders"*		
10	Case 7	*"one of [the founders] has his own fields in Grasse. [...] Reflecting on how he operated in alignment with sustainable development, harvesting himself [...] the idea [...] came directly from the founders"*		
11	Case 9	*"I simply could not imagine delivering any other way than by bike"*		
12	Case 9	*"All the raw materials are sourced from farms around my home. [...] I had solar panels installed and I produce during the day to optimise production"*		
13	Case 9	*"I am from Eastern France and I did all my internships in Germany when I was young. The ecological awareness in Germany was very, very well developed"*		
14	Case 11	*"We are committed to sustainable and responsible fashion. [...] this goes through every daily action"*		
15	Case 11	*"From the very launch of the company, we made the deliberate choice to offer free delivery on every order, because this allows us to choose the delivery method ourselves"*		
16	Case 1	*"Customer obsession currently means pursuing a greener delivery where possible"*		
17	Case 1	*"No, I think it's a genuine internal driver"*		
18	Case 10	*"We call ourselves [Anonymized], there is a strong commitment to staying local, artisanal and French"*		
19	Case 1	*"Environmental questions are at the centre of the budget decisions we make"*		
20	Case 1	*"it's also visible in the rest of the Supply Chain, where we have several internal projects to reduce emissions across the entire distribution chain. I'm thinking particularly of the internalisation of customer returns [...] we are trying to move towards eliminating the Middle Mile, the Sort Center, in order to further reduce our transport costs and the emissions associated with transportation"*		
21	Case 1	*"we also have photovoltaic installations at several of our sites that help us reduce fuel consumption"*		
22	Case 10	*"We consider ourselves a responsible company, we collaborate with associations like the [Anonymized]. Everything that cannot be sold because it has passed its best-before date, the DLU is donated."*		
23	Case 10	*"We also work with ESAT establishments, facilities that employ people with disabilities, for all our copacking work."*		
24	Case 10	*"We have a deep CSR commitment. So the ecological and sustainable dimension is part of those values. We try to develop it more and more, even if it is not always simple"*		
25	Case 11	*"We are aware that the fashion world is an ecological disaster and we do not want to add to it. Everything is thought through to reduce our carbon footprint as much as possible, from sourcing, procurement and production of raw materials, to the emissions of our pop-up shops"*		
26	Case 11	*Concretely, we have chosen a short-circuit approach through local production in France, recycled materials and products designed to last. We have banned air transport for all our products and materials, as it is the most polluting mode of transport"*		
27	Case 11	*"In terms of restocking, we adapt production to sales, which avoids overstocking and reduces unsold inventory"*		
28	Case 11	*"Having that choice on our side means we can select the most ecological delivery option available from the carrier, even if it costs us more [...] We work with UPS and Colissimo, and we decarbonise all our shipments. Even if it costs more, it is essential"*	**Founding values and brand identity alignment**	
29	Case 2	*"We wanted to work with partners who were aligned with us"*		
30	Case 2	*"it is genuinely a source of pride to represent the company and its values, and to see that the customer demand is met and that customers are happy and satisfied, both in terms of their food being local and genuinely a short, closed-loop circuit"*		
31	Case 3	*"we have a dedicated CSR unit and a transport director who is strongly aligned on the subject"*		
32	Case 3	*"we have had someone responsible for CSR for a long time. At the beginning it was more global QHSE. But for the last 15 years, there has been a dedicated person. And the company is 20 years old"*		
33	Case 5	*"It is a brand that positions itself as quite green, I would say. We are not in the low-end segment in terms of price or product formulation. So it is really a clean premium brand that is committed to beauty and also to health more generally"*		
34	Case 5	*"there is still a real margin to make the whole brand green, truly sustainable. We are really more focused on the products than on the delivery, and there is still a lot of work to make that genuinely visible"*		
35	Case 5	*"I would tell them to go for it. Obviously you need the funds, you need investors behind you. But you should go for it because it will immediately be in line with everything we want to stand for today and tomorrow"*		
36	Case 6	*We source books within a 150-kilometre radius around the packing zone, that was quite important to us"*		
37	Case 6	*"No, it was included from the start"*		
38	Case 6	*"We make it a central element of the business model and we communicate a lot around it: it is part of the DNA of the company"*		
39	Case 6	*"We work with an ESAT, a work assistance establishment for people with disabilities, which handles reception of books, checking, cleaning, repair if needed, and then monthly packing."*		
40	Case 7	*"This is a process that was put in place from the very beginning of [Anonymized], to be aligned with the vision of the house"*		
41	Case 7	*"simply on the principle that we wanted to stay aligned with the core idea of the house. Contractually, we went directly to look for prospects who could deliver on that"*		
42	Case 7	*"For the remaining products, sourcing may happen abroad, but it is always sourced in correlation with sustainable development and fair trade practices, to remain in line with the company's respectful approach to the environment"*		
43	Case 7	*"if we want to continue preserving the planet and remain aligned with the mindset of consumers, it is something that needs to be put in place, otherwise you are putting obstacles in your own way"*		
44	Case 8	*"I would tend to say it is still a topic that comes up regularly. There are also training sessions on the subject. So we are sensitive to it"*		
45	Case 8	*"I would say it is chosen. We are not forced, these are decisions made by the company, even if they create challenges"*		
46	Case 9	*"I try to offer this broth to people within a radius around my home, in order to be a hyper-responsible mini-enterprise, which is consistent with my values"*		**Normative Drivers**
47	Case 9	*"It was not even a question. I could not imagine making a super virtuous broth and delivering it with an anti-virtuous solution. It simply did not fit the company's values at all"*		
48	Case 9	*"The bottles are glass bottles that are obviously recycled at each production"*		
49	Case 9	*"I also installed a reverse osmosis water system to have very pure water and avoid back-and-forth trips buying bottled water. All the herbs come from my kitchen garden"*		

	A	B	C	D
50	Case 9	*"I think you need to really be in alignment with your values. And I would suggest asking the founder about their responsibility as a human being, for their own health first, and for the health of the environment, so that they can consciously embed that responsibility into their policies and actions"*		
51	Case 9	*"We are not alone on this planet and we all need to be responsible for our lives, and to leave a livable environment for our children"*		
52	Case 4	*"Was not really planned from the start, even if being able to get a label and have an additional argument to use [Anonymised] was an objective. But it was not the primary motivation, it was fundamentally based on our values"*		
53	Case 2	*"The company was created around a demand from customers to eat better"*		
54	Case 4	*"There are clients who would not be particularly enthusiastic about delivery if it didn't align with their ethics or their image, if the crockery was new, bought from a wholesaler, or delivered by a big diesel truck"*		
55	Case 5	*"We have a clientele that is relatively affluent in terms of what they buy, but also very interested in sustainable development and in really paying attention to what they put on their nails, and also what that generates downstream"*		
56	Case 7	*"the consumer's gaze regarding our responsibility in this space. I think that helped a lot in putting this in place"*		
57	Case 7	*"I think we would lose a certain segment of the population, on the principle of proposing something and then discontinuing it"*		
58	Case 8	*"The customer today is not very demanding on this type of topic. What they mainly want is to receive their goods as quickly as possible and without damage"*		
59	Case 10	*"Our overall brand values resonate with theirs. The way we deliver is part of that broader picture [...] not the primary selling point in isolation"*		
60	Case 1	*"there is obviously a growing environmental awareness among our customers. And again, 'Customer Obsession' has always been one of [Anonymized]'s core beliefs, it's what we call a 'Leadership Principle', a management principle at [Anonymized] that was developed by [Anonymized] at the time"*		
61	Case 4	*Yes, I think it is valued because we get talked about, people mention us because of the electric van and we get a lot of recommendations because of it.*		
62	Case 11	*"If our discourse is disconnected from our actions, then yes, I think it would affect sales, because it is hard to inspire trust in customers that way"*		
63	Case 11	*"One problem we identified is that when delivery is not included in the price, customers tend to choose the least ecological delivery option because it is the cheapest"*		
64	Case 2	*"We ran a survey, a panel of about 1,000 customers, and we asked them, for example, if the price of a tomato were to increase, how much would they be willing to pay more? And honestly, I believe we had around 75% positive responses"*	**Customer expectations and trust**	
65	Case 2	*"customers today genuinely want that eco-responsible, sustainable side. And you need to be careful not to confuse this with greenwashing"*		
66	Case 3	*"Either the brands are very aware of CSR, in which case there will be a CSR impact on the last mile. Or the challenge is related to goods security"*		
67	Case 3	*"one of our clients, [Anonymized], is very focused on CSR because it is part of their brand DNA. So when we are looking for a transport solution for them, we obviously ask ourselves how we can make those last-mile operations as green as possible."*		
68	Case 5	*"customers, even if they consider the product to be good for health, do not necessarily see the other side of things"*		
69	Case 7	*"I think there would be different types of customers who would position themselves differently on this. I think some could be sceptical if we were to stop it"*		
70	Case 7	*"we are also a company positioned in the luxury segment, and luxury is not the sector most aligned with sustainable development, or at least with implementing solutions that go with it"*		
71	Case 7	*"it can sometimes be difficult to manage from the consumer's perspective, they want something luxurious but also sustainable. And it is genuinely complicated to combine the two at certain moments, between what the consumer wants and what is factually possible"*		
72	Case 8	*"Today, green logistics is difficult mainly because customers expect speed. When you click, you want delivery the next day or immediately"*		
73	Case 8	*"If customers accepted longer delivery times, more sustainable solutions would be possible. This is why offering delivery choices is important"*		
74	Case 9	*"people are very receptive when you set the example. They are very happy to see there are other solutions"*		
75	Case 3	*"I think societal change would be number one. Today, whether in personal or professional life, everyone is sensitised to the topic"*		
76	Case 3	*"Future employees and company directors will have it built into their habits"*		
77	Case 3	*"We are based near Saint-Malo lots of sailors, people who love nature. So naturally there is an attraction to the topic"*		
78	Case 7	*"It has become a less taboo subject, companies are talking about it more"*		
79	Case 4	*"I think it is necessary that everyone adopts this kind of solution"*		
80	Case 2	*"Customers today are done with [greenwashing]. They are coherent, they can tell the difference between greenwashing and something real"*		
81	Case 11	*"We saw during the 2020 lockdown that the sky over Paris had never been so clear, due to the fact that traffic was almost nonexistent. I think that awakened awareness"*	**Societal evolution and professional norms**	
82	Case 11	*"I think it is essential today to adopt sustainable solutions, whatever the reasons. As long as there is no greenwashing, something I cannot stand, everyone can find their benefit in adopting sustainable solutions"*		
83	Case 11	*People realise that if we do not act now, the consequences could be irreversible"*		
84	Case 11	*"there was a kind of momentum to make sustainable solutions easier to implement, since people were demanding them"*		
85	Case 2	*"I hope this will open up this path toward more sustainable business creation across the board"*		
86	Case 7	*"Sixteen years ago, the discourse was less open and actions were less engaged on sustainable development"*		
87	Case 7	*"in today's world, it is hard to really ignore this and put blinders on. It is something you have to move forward with now"*		
88	Case 9	*"Honestly, it is moving very slowly. Today we sort our rubbish well, in Germany that was already the case 30 years ago"*		
89	Case 6	*"A differentiating and exclusionary criterion relative to the other offers on the market"*		
90	Case 6	*"The ecological criterion, not just on delivery but more broadly on the choice of business model was indeed somewhat motivated by a desire for differentiation relative to those competitors"*		
91	Case 7	*"It is really about standing out from competitors"*		
92	Case 4	*"Coming back to our brand image and the perception people have when they come to us, it can be a real advantage. And no one is against ecology and reducing emissions"*		
93	Case 11	*"When we were researching during the creation of our brand, we noticed that clothing brands do not communicate much about last-mile sustainability. [...] So yes, it influenced us, because we differentiate ourselves already through our delivery method alone"*		
94	Case 3	*"It is definitely part of the advantages you need to have to measure the business impact"*		
95	Case 3	*"The big players on the market have green solutions too. It is becoming the norm"*		
96	Case 1	*"I would consider us more as pioneers, given the scale of our operations"*		

	A	B	C	D
97	Case 9	*"I think the solution is always influenced by a person through their own example and exemplarity. I think I touch people through the exemplarity of this way of offering an absolutely wonderful product in a hyper-ecological way"*	**Sustainability as competitive advantage**	**Competitive and Economic Drivers**
98	Case 4	*the company and the van helped us to obtain an eco-responsible label, which also gives us a very professional image and builds a lot of trust with our clients.*		
99	Case 11	*"Yes, today many online stores are considering sustainable deliveries, but I think very few of us actually offer free delivery specifically in order to genuinely decarbonise it"*		
100	Case 11	*"All the effort seems to go into recyclable and recycled packaging, but for us that was not enough"*		
101	Case 2	*"you should not wait. We were a little bit among the first on the market to do this. And the demand is genuinely there"*		
102	Case 2	*"My main piece of advice is really to just go for it, not to keep waiting behind the big players who are ultimately more focused on performance and their budgets and much less on the real, concrete demand from customers"*		
103	Case 3	*"This allows the company to obtain labels like the EcoVadis label. We very recently renewed and obtained the Silver Medal"*		
104	Case 3	*"from a communication standpoint for brands that work with us or come to us for business, it positions where [Anonymized] stands on this topic"*		
105	Case 5	*it truly differentiates us, in terms of our formulation, our storytelling. And now it is true that customers are much more oriented towards sustainable development"*		
106	Case 5	*"given that we have good market leadership, at least in the green nail beauty sector, which is still relatively niche, there was no need to align with competitors"*		
107	Case 6	*"it is also a communication lever in a CSR and ecological policy, riding the wave that is on the one hand fashionable, and on the other hand gives the company additional communication arguments"*		
108	Case 8	*"Let's say we are the leaders in the sector, but that does not prevent us from taking inspiration from our competitors"*		
109	Case 9	*"one of my clients recently bought a bike to cycle to work, because they realised they worked 5 km from home and that there might be an ecological reason to cycle. So once again, it is a story of exemplarity"*		
110	Case 1	*"Last Mile delivery accounts for 70 to 80% of the total cost of delivering a product. [...] micromobility and the [Electric van] allow us to deliver more parcels more often at a lower cost"*	**Cost optimisation through sustainability**	
111	Case 2	*"We are very happy about that, especially given the rising price of petrol"*		
112	Case 2	*"Sustainable delivery is more expensive. That was another challenge we faced, but one we managed to overcome"*		
113	Case 4	*"What also motivated us a lot at the start was fuel cost. Doing 300 km in a single day, at some point that adds up quite a bit. So that is partly why we did not really hesitate and invested directly in an electric van"*		
114	Case 6	*"These solutions are interesting from several perspectives, both financially for the company and for the customer, in the sense that the final price the customer pays is lower than home delivery"*		
115	Case 10	*"To be honest, one of the first motivations is cost. By consolidating and optimising routes, we limit the costs of temperature-controlled delivery, which is very expensive due to the energy requirements"*		
116	Case 3	*"Do you want to keep your business alive, or do you want to invest in CSR? The question can be asked today. I don't feel that CSR is something that is prioritised over survival"*		
117	Case 8	*"The cost of an electric truck is extremely high. So we have to calculate how we are going to amortise this cost over 2 to 3 years"*		
118	Case 1	*"For an SME, there are definitely financial questions, that's obvious. And again, the investment made in the [Electric van] is something very few companies are doing"*		
119	Case 1	*"the fact that oil prices risk exploding due to tensions in the Middle East is a factor for all logistics chains worldwide. And we are glad to have taken the lead and already implemented several electric and micromobility solutions to reduce these dependencies on oil"*		
120	Case 10	*"The recent geopolitical context with the conflict in Iran has pushed up transport costs significantly, which is forcing us to rethink"*		
121	Case 10	*"We also try to avoid empty trips, if I send a carrier to a client in Boulogne, I try to organise a collection in that same area for the return journey. It is both more ecological and more cost-effective"*		
122	Case 10	*"Temperature-controlled delivery involves large refrigerated trucks, which are both costly and more polluting"*		
123	Case 2	*"We ran two changes: now when we make a delivery, we collect their previous jars at the same time. [...] We have grouped them, we do rounds every two days per drive point. [...] I think today things have really changed, I believe we gained around 10% in performance because of these two changes"*		
124	Case 2	*"On the budget side, it is also worth looking at what the State can offer, because sometimes there is financial support available. So if budget is the only obstacle, there are always solutions"*		
125	Case 3	*"if on the last mile we can anticipate by using electric vehicles or biofuels, it helps to absorb a bit of those cost surges"*		
126	Case 3	*"companies are under pressure on their transport costs. They want to pull costs down. If a company doesn't have the cash flow or the budget margin to invest in this, they will prefer to cut costs"*		
127	Case 5	*"we do not necessarily want our customers to feel the difference, so it tends to be managed internally"*		
128	Case 6	*"it allows for lower prices, which today is a strong argument given inflation and the state of the economy where we need to stimulate consumption through lower prices. So that is an advantage for both the customer and the company"*		
129	Case 6	*By including delivery in the price, the delivery becomes 'offered' rather than free. And it is important to say it is offered rather than free, in terms of messaging, free makes delivery seem cheap, whereas offered makes it feel like an additional service provided by the company"*		
130	Case 6	*"So by including delivery in the box price, customers by default choose the relay-point option"*		
131	Case 7	*"No, it is included in the delivery cost. [...] It is something that is absorbed by the company, through the negotiations we conducted with the partner we work with on this"*		
132	Case 7	*"It can feel scary initially from a budget perspective, but in the end the costs are relatively absorbed"*		
133	Case 8	*"often in companies, when a project is launched, large companies require a return on investment within 3 years. So the analysis was done in that sense"*		
134	Case 2	*"No, it was not really regulation."*	**Direct regulatory influence (weak)**	**Direct regulatory influence (weak)**
135	Case 2	*"That is more of a minor factor"*		
136	Case 4	*"Even without any rules, we would have adopted the electric van anyway"*		
137	Case 6	*"Not at our level, at least at my level, we do not feel them much"*		
138	Case 9	*"I prefer to be small, autonomous, responsible and ecological on my own, and make the government bend eventually because it is logical"*		
139	Case 11	*"We are not large enough to be directly concerned by government regulations"*		
140	Case 1	*"I know we are also in partnership with the city of Munich. We are obviously in constant contact with local officials to find the broadest possible solutions, with a favourable tax framework and integration into the local landscape"*		
141	Case 8	*"regarding regulations, yes, they must be taken into account and read, but today that is mainly the role of carriers, not ours"*		
142	Case 9	*"I am not at all impacted by government laws, it is above my head"*		
143	Case 10	*"We want to be as clean as possible in those documents so that sustainability requirements do not block future commercial relationships"*		
144	Case 3	*"Tomorrow, the idea is not to be caught off guard by government restrictions, we have the solution."*		
145	Case 11	*"We are ambitious and obviously want to grow as much as possible. So given the possibility that at some point [...] we could be concerned by these laws, yes, we are glad that we are already aligned"*		
146	Case 10	*More and more corporate clients will require us to go through that kind of platform. [...] So it is anticipatory, we are trying to get ahead of regulations that may tighten in the future, even if they do not directly apply to us today"*		

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147	Case 11	*"In any case, given what has been happening over the past few years, it would not be surprising if in the fairly near future sustainable deliveries became mandatory for everyone. It is therefore better to get on the train now"*	**Anticipatory regulatory compliance**	**Coercive and Contextual Pressures**
148	Case 3	*"we have the benefit of working with a huge number of partners and our list of available services is very long. [...] what [Anonymized] does to prepare for tomorrow, meaning stricter restrictions, is to have backup solutions ready to activate"*		
149	Case 3	*"If a client comes to us and says 'OK, I have no choice in this country, I have to deliver this way', no problem, we have the right solutions, we'll activate them. It's about anticipation, having several strings to your bow"*		
150	Case 8	*"On the France network, I think it is through that approach that we challenge our carriers. That's why we ask them what they are implementing, to be ahead of major deadlines like 2030 or 2050"*		
151	Case 10	*"We use a B2B platform called [Anonymised] where we have to fill in all company information including a full RSE section"*		
152	Case 8	*"it is our role within transport and also sometimes sourcing, so procurement, to challenge carriers to obtain information on what is being implemented"*		
153	Case 8	*"Carriers, on their side, know that there are more and more demands from governments and regulations on these topics. So they either have to take the lead or comply when regulations are implemented"*	**Delegated coercive pressure (carriers)**	
154	Case 8	*"Today that is mainly the role of carriers, not ours, since we do not own transport operations"*		
155	Case 6	*"The transport side is really the carrier's problem, it is their responsibility to figure out how they are going to route the parcel through the city centre"*		
156	Case 1	*"we have a defined set of vehicle types that we offer to our DSPs, Delivery Service Partners [...] So at certain Delivery Stations we only use [Electric van], and in that case we provide them to our delivery partners"*		
157	Case 11	*"All these carriers are increasingly implementing solutions to decarbonise their routes: electric trucks, cargo bikes, relay points. Although this is much more developed in large cities, I believe that in the coming years it will be possible to deliver everything by a decarbonised method"*		
158	Case 3	*"a carrier who tells us 'I deliver Paris and all the major cities by electric vehicle or low-emission vehicle', we go with them. If a carrier tells us 'To offset my CO2 emissions, I plant trees in the Amazon', I am a bit less keen"*		
159	Case 5	*"Since the offices are in central Paris, there is a lot of passage, a lot of routes, a lot of cycle lanes. For cargo bikes, it is not complicated to come and collect the goods from the boutique and deliver them to customers"*	**Urban infrastructure as enabler**	
160	Case 9	*"The last mayoral term was really focused on developing cycle paths. So honestly, everything is set up to make the cargo bike solution very easy to implement"*		
161	Case 11	*"In general, the denser the urban context, the more sustainable delivery solutions are available"*		
162	Case 8	*"There were many restricted zones, red zones and what the city did was create dedicated parking areas for delivery drivers. [...] It was an ecological approach and it worked very well"*		
163	Case 9	*"Around me, we now have many zones limited to 30 km/h and we can see cycling genuinely developing. It is extraordinary"*		
164	Case 10	*"Paris has historically been and I think will remain a city that prioritises reducing traffic"*		
165	Case 1	*"in Munich there is also a strong environmental awareness and in recent years we've really seen the rate of [Electric van] explode"*		
166	Case 10	*"In that context, where there is a political will to reduce road traffic, it forces us to think about how to evolve. [...] yes, this context pushes us to reflect and adapt, and I would say it is very much linked to local political choices and municipal elections"*		
167	Case 11	*"In Paris again, the solutions are very numerous and I think cargo bikes are going to take a really significant share of the Parisian market in the coming years. When you walk around the city, you can see that bikes move much more fluidly than vans. They pollute less and are far more flexible than other vehicles"*		
168	Case 2	*"The city is even supporting us financially in that regard. [...] I think we are genuinely confident about the future. As I said, the city is behind us and with us"*		
169	Case 2	*"We are also seeing customers shift, even in terms of their own car choices. So that will continue to develop on that front"*		
170	Case 3	*"for these clients who are very sensitive to CSR and last-mile green solutions, I think between 50 and 60% of their deliveries can actually be green"*		
171	Case 3	*"I think it can go a bit higher and we can be closer to around 60–70% when we're talking about metropolises and major cities"*		
172	Case 6	*"Mondial Relay has more relay points than the others. So we went with Mondial Relay"*		
173	Case 6	*"There is also a practical side to relay-point delivery, the parcel stays at the point for a while, which gives customers flexibility to pick it up when they want"*		
174	Case 7	*"In central Paris, it is faster"*		
175	Case 8	*"some cities are more suitable for implementing bike deliveries because they have bike lanes, because the routes are easier to optimize, or because they are smaller cities where local agencies are closer to the city center, making round trips easier"*		
176	Case 8	*"In the future, having dedicated delivery lanes, like taxi lanes, could improve efficiency"*		
177	Case 8	*"we are lucky in the sense that we do not need large trucks, since everything is done with vans or small delivery vehicles, which allows us to move around Paris as efficiently as possible"*		
178	Case 8	*"we also have a major advantage, which is that the majority of our network operates at night. So we do a lot of deliveries to opticians via mailboxes, which allows us to deposit goods. That means we can drive faster because there are fewer people on the road"*		
179	Case 9	*"The cycle lanes are wider. They serve almost the whole town. We even have paths outside the roads"*		
180	Case 9	*"yes, we have benefited from the ecological momentum of our town"*		
181	Case 9	*"We can feel we are getting closer to the German model, we will end up with more bikes than cars. And that is wonderful, because even in terms of sound, you can hear the birds now"*		
182	Case 1	*"The [Electric van] are larger than conventional vans. [...] I'm thinking for example of Amsterdam, where we carry out a very large share of our deliveries via micromobility because the vans simply cannot pass through most of the streets"*	**Urban context as constraint**	
183	Case 1	*"These are very tall vans. So you need to find specific parking spots to park them, and you need to make sure you're going through areas where they won't obstruct public space"*		
184	Case 10	*"Paris is genuinely very complex logistically. Every time we need a delivery, there is a risk of getting stuck in traffic. [...] clients often have strict delivery bay restrictions, which means I have to provide the driver's identity and vehicle registration number in advance"*		
185	Case 8	*"You can carry many more parcels in a van than on a bike, so we need to have the ability to go back and forth to the sorting centre or the local agency in order to maintain the same delivery standards as with vans"*		
186	Case 4	*"Our goal is to be able to deliver everywhere, within the limits of our range, obviously, and to not have to say no to future clients who might be located in a restricted area"*		
187	Case 7	*"I think unconsciously, in any case, part of the reflection was inevitably based on that as well, because you have to respect what is happening around you and the life of the city in which you operate."*		
188	Case 11	*"When we were implementing our sustainable solutions, the urban context did not play an important role in our decision. However, today I realise that it actually plays a significant role in the decarbonisation of our shipments"*		
189	Case 1	*"the size of the US vans, given local infrastructure, was much larger than most European cities could accommodate. So we had to revise their development to reduce their volume so they could fit through most of our routes"*		
190	Case 1	*"the local climate, local infrastructure, local urban planning, these are key factors for implementing these delivery solutions"*		

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191	Case 10	*In summer, it is simply impossible to ship chocolate via a standard carrier like Colissimo, because if a parcel spends two or three days in transit, possibly sitting in places where temperatures can reach 26 to 35 degrees, it is absolutely impossible to leave it at ambient temperature"*		
192	Case 3	*"If it's a small town, somewhere in the middle of nowhere, it's a bit more complicated"*		
193	Case 3	*"our warehouses are not in city centres, it is harder to deliver goods by low-emission vehicle back to the warehouse. So finding sustainable return solutions is more complex"*		
194	Case 7	*"all our teams, sites and production facilities are based in Paris, which is what makes cargo bike delivery feasible there. But expanding it to the rest of France is a long-term project"*		
195	Case 7	*"in a city that is large but not as dense and urbanised as Paris, electric vehicles and other solutions beyond cargo bikes could be viable"*		
196	Case 7	*"The example of Lille, the fact that cars are no longer accepted in the city centre, does give food for thought on the principle that we will inevitably want to align with how the city is thinking"*		
197	Case 9	*"I admit that if we go beyond about 20 km, an electric vehicle solution would become viable, because you also have to be practical. You cannot spend four hours delivering by bike"*		
198	Case 9	*"the production needs to be delivered within a time window that corresponds to people getting home from work"*		
199	Case 4	*"We don't really know any direct competitors"*		
200	Case 7	*"I am not sure. Personally, I do not know many similar cases"*		
201	Case 5	*"That was a fairly minor aspect, because honestly we already had very strong drivers before we even had to worry about competitors"*		
202	Case 3	*"I honestly couldn't say yes or no. [...] I would say it was really baked into the DNA from the start"*		
203	Case 2	*"We did not really focus on what was around us in the region"*		
204	Case 1	*"for all companies there is inevitably a kind of cascade effect that has impacted us"*		
205	Case 11	*"Although awareness was lower at the time, I think we would still have adopted the principle of offering free delivery so that we could choose the most sustainable delivery method ourselves"*	**Limited direct peer imitation**	
206	Case 2	*"In terms of direct comparisons, [Anonymized] operates on a somewhat similar model with a centralised hub. But the idea of truly putting customers at the centre of operations, I have not seen that much elsewhere"*		
207	Case 2	*"We tend to do that now, but honestly at the beginning, no"*		
208	Case 7	*"especially when we started thinking about it during Covid, it was not something that was very widespread at the time"*		
209	Case 9	*"Honestly, yes a definite yes. [...] 10 years ago I would have done exactly the same thing"*		
210	Case 10	*"We drew inspiration not from direct competitors but from wine distributors who share our logistics provider [Anonymized]. They face the same temperature and handling challenges we do"*		**Mimetic Dynamics**
211	Case 5	*"We looked at brands simply positioned in the sustainable development space more broadly"*		
212	Case 5	*"We also looked at what other brands were doing, not so much in the beauty sector specifically but brands simply positioned in the sustainable development space more broadly. We saw how they operated, and then we just looked, compared, and figured out how it could work with electric bikes and cargo bikes"*		
213	Case 3	*"management attended external trade shows and events where companies come together, so there must have been some influence"*	**Indirect or cross-sector mimicry**	
214	Case 3	*"a direct influence at the operational level or at the level of middle managers, we don't have that information"*		
215	Case 8	*"for daytime delivery, however, there are always new players entering the market, and it is possible for us to integrate them or observe what they do and see if it can be proposed to our current carriers"*		
216	Case 8	*"that does not prevent us from taking inspiration from our competitors"*		
217	Case 9	*"There is one company that comes to mind straight away, not with a cargo bike, but they use a small electric van. [...] yes, one is called [Anonymized] and the other is [Anonymized]. Yes, we can feel that things are moving and there is a real will to be very responsible"*		
218	Case 6	*"It is true that the ecological criterion, not just on delivery but more broadly on the choice of business model, was indeed somewhat motivated by a desire for differentiation relative to those competitors"*		
219	Case 6	*"I would put competition, customers, and ecological awareness in first place, roughly in that order"*		
220	Case 6	*"Taking into account that home delivery was cheaper [ten years ago], I probably would have prioritised it a bit less for economic reasons. And the ecological aspect, customers were also less concerned about it"*		
221	Case 6	*"We do not really have direct competitors on the territory. We have a few competitors, at least in the Paca region, but mostly at the national level, based in Paris or other cities"*	**Competitive field awareness**	
222	Case 4	*"Not at all especially not to competitors. You don't give advice to competitors"*		
223	Case 4	*"I think yes, because we get a lot of feedback from our service providers telling us it's great, not exactly encouraging us, but making them think: 'Oh yes, maybe that would be a good idea.' I am not saying at all that we are an example, but I think many people are asking themselves why they wouldn't go electric"*		
224	Case 11	*"Yes, today many online stores are considering sustainable deliveries, but I think very few of us actually offer free delivery specifically in order to genuinely decarbonise it"*		

Appendix 3: Declaration - AI usage

We hereby declare that this thesis represents our own work.

We declare that we have obtained the necessary ethical approval (e.g., GDPR) and acknowledged our obligations and the rights of the participants/respondents in the research.
We declare that if we have made use of non-sensitive personal data we have collected *GDPR Thesis consent forms*, including contact details with participants in our study.
We have read and applied the Jönköping University's current research ethics guidelines with regards to the use of artificial intelligence (AI) tool in this work as presented in the Course Guide. During the preparation of this work the authors used the following AI tools and the purpose for their use:

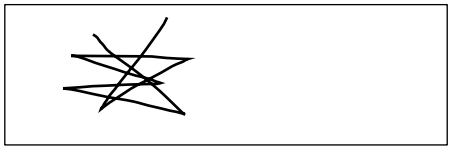
- Claude (Anthropic) to translate the interviews from French to English.
- Claude (Anthropic) to improve the text by translating, finding grammatical and coherence errors and polishing it to ensure the best level of academic English possible.
- Google transcription tool to transcribe the interviews via the Google meet tool.
- Claude (Anthropic) to provide critiques and feedback on draft sections, which the authors critically evaluated before manually revising their ideas.

As authors, we have reviewed and edited the content as needed and take full responsibility for the content of the thesis.

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